

Resource limitation rewires chromosome instability and ploidy evolution across *in vitro* and *in vivo* cancer models

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Abstract

Chromosome instability (CIN) generates karyotypic diversity, but whether altered progeny persist may depend on resource availability and ploidy. We combined lineage-matched near-diploid and near-tetraploid SUM159 culture and tumor datasets with a chromosome-number-structured mechanistic model linking oxygen/resource stress to proliferation, death, whole-genome doubling, chromosome missegregation, and post-missegregation survival. *In vitro*, oxygen deprivation reduced passage growth and remodeled low- and high-chromosome components. The model reproduced the emergence and expansion of high-chromosome states and complementary ploidy shifts from 2N and 4N starting states. Direct hypoxia-origin death in the 4N lineage remained below 1.9% across the ten lowest-objective fits, although the model did not recover the distinct terminal mixture proportions of parallel oxygen-deprived 2N lineages. Separate *in vivo* fits captured cohort-scale tumor growth and ploidy remodeling but admitted multiple near-equivalent parameterizations. Across six warm-start-conditioned joint-fit ensembles, joint fits consistently estimated a lower maximal division rate *in vivo* and higher complete effective per-chromosome missegregation probabilities at 0% and 1% O₂. Complete post-missegregation survival functions revealed a two-part context contrast: altered daughters had higher absolute fitted survival *in vivo* at both 44 and 88 chromosomes, whereas the survival increase between these states was larger *in vitro*. Fixed-O₂ attractor analyses retained multiple compatible response regimes, indicating that the global oxygen–CIN–ploidy landscape was not uniquely resolved. Together, these results support a model in which resource context changes both the production of chromosome-number variation and the ploidy-dependent filtering of viable descendants. These conclusions remain conditional on the model, data streams, parameter bounds, regularization, and sampled optimizer landscapes.

Introduction

Whole-genome doubling (WGD) and aneuploidy are common features of cancer evolution, yet their prevalence varies substantially across biological contexts.^{1–3} A central unresolved question is why high-ploidy states are tolerated in some environments but selected against in others.^{4,5} One clue

comes from the striking difference between cultured cancer cell lines and primary tumors: across organ systems, cell lines exhibit markedly higher ploidy than primary tumors. This observation suggests that standard in vitro culture conditions, which provide abundant nutrients and simplified environmental constraints, may relax selection against high-ploidy states, whereas in vivo tumor environments impose stronger constraints on cells carrying excess genomic material.^{4,5}

A plausible basis for this context dependence is the energetic cost of increased chromosome content.⁴⁻⁶ WGD increases the biosynthetic burden associated with DNA replication, RNA production, and especially protein synthesis.⁴⁻⁶ Aneuploidy further compounds this cost by disrupting gene dosage balance, producing proteotoxic, osmotic, oxidative, replication, and endoplasmic-reticulum stress.⁷⁻⁹ These stresses require active homeostatic responses, including protein quality control, degradation of excess proteins, DNA repair, and metabolic adaptation.^{5,7-9} Thus, high-ploidy cells are expected to carry a resource-dependent fitness cost, particularly when oxygen, glucose, or other metabolic substrates become limiting.^{4,5}

At the same time, resource scarcity can also increase genomic instability.^{4,5} Environmental stresses such as serum starvation, hypoxia, folate deficiency, and energy restriction have been linked to karyotypic instability, chromosome mis-segregation, polyploidization, and increased cellular heterogeneity. The evolutionary consequences of these events depend strongly on ploidy. Chromosome mis-segregation can trigger p53-mediated arrest, senescence, or death, particularly when chromosome loss creates severe dosage imbalance.^{8,10-13} WGD and high-ploidy states can buffer these effects by reducing the relative dosage impact of individual chromosome gains or losses, allowing some mis-segregations to be converted into heritable karyotype variation rather than lethal events.^{7,14,15} Consistent with this view, ALFA-K, a computational method that infers chromosome-level karyotype fitness from longitudinal single-cell karyotype data, showed that mis-segregations impose stronger fitness costs near diploidy than near tetraploidy.¹⁶ Thus, ploidy is a quantitative determinant of whether CIN generates nonviable progeny or heritable karyotypic diversity.

Oxygen provides a tractable first-order proxy for this resource landscape.⁴⁻⁶ Across tissue sites, WGD prevalence correlates with peripheral oxygen partial pressure, consistent with the idea that well-oxygenated tissues are more permissive for WGD-positive cancers, whereas poorly oxygenated tissues are less permissive.^{5,6} However, oxygen is an imperfect surrogate for total energy availability. Tissue resource states also depend on amino acid and glucose availability, local metabolic consumption, and metabolic specialization. For example, tissues that rely heavily on glycolysis or have unusually high glucose availability may not experience low oxygen as equivalent to global energy scarcity.

Together, these observations identify a central tension in cancer ploidy evolution. Resource scarcity should select against WGD and other high-ploidy states because additional chromosome content increases biosynthetic, proteostatic, and homeostatic demands.⁴⁻⁶ Yet adverse environments can also promote CIN, chromosome mis-segregation, and polyploidization, generating the karyotypic variation on which selection acts.^{4,5} Whether this variation is eliminated or retained depends in part on ploidy: near-diploid cells are more vulnerable to lethal dosage imbalance after chromosome loss, whereas WGD or high-ploidy cells can better buffer individual segregation errors and convert them into heritable karyotype variation.^{7,14-16} Thus, resource limitation may simultaneously oppose and promote WGD, depending on how strongly it suppresses proliferation, induces CIN, and filters the survival of altered progeny. A

key question is therefore which environments select for WGD, which select against it, and through which mechanisms. Answering this requires a quantitative framework that can jointly represent resource-dependent proliferation, stress-induced chromosome instability, and ploidy-dependent survival of resulting karyotypic states.

The central analytical target of this study is therefore not a single fitted parameter, but the context-dependent mapping from resource stress to chromosome-number variation and from altered daughters to viable diversity. By fitting one mechanistic structure to lineage-matched SUM159 culture and tumor data, we asked which complete response functions showed concordant context contrasts, which observations the model failed to reproduce, and which mechanistic interpretations remained unresolved.

Methods

Study design and data streams. We analyzed near-diploid and near-tetraploid SUM159 lineages across two experimental contexts. *In vivo*, orthotopic tumors initiated from the near-diploid and near-tetraploid lineages were followed with longitudinal tumor-burden measurements and harvested for terminal single-cell chromosome-number profiling; where matched terminal histology was available, necrosis was quantified at harvest. The *in vivo* calibration used the untreated subset of eight tumors, four initiated from each lineage. *In vitro*, the corresponding lineage pair was analyzed in passage-structured culture experiments with measured passage growth, karyotype/chromosome-number observations, and G0/G1 flow-density profiles under controlled oxygen conditions. The shared lineage origins placed the two datasets on a common modeling axis, but the experiments were not biological replicates of one another and did not constitute a randomized intervention comparing culture with tumors.

Model state and oxygen context. The model represents viable cells as a distribution over total autosomal chromosome-number states. It also tracks non-viable biomass in separate dead compartments, so simulated bulk tumor burden includes both live cells and uncleared dead material. Terminal single-cell chromosome-number profiles sample only the viable population at harvest, whereas tumor burden and necrotic fraction depend on live plus uncleared dead compartments. This state representation lets one simulation be compared simultaneously with longitudinal tumor volume, terminal necrosis, and single-cell chromosome-number distributions, while preserving the model’s focus on karyotype evolution.

Resource context is represented differently in the two settings. *In vivo*, oxygen is modeled as an effective mean-field level that declines with chromosome-weighted tumor demand and relaxes toward a supply-demand target. *In vitro*, each passage segment is simulated at its assigned fixed oxygen level.

Mechanistic modules. The mechanistic core of the model has three modules corresponding to the routes through which resource context could select for or against WGD. First, oxygen/resource stress suppresses proliferation through a chromosome-number-dependent growth penalty, so high-chromosome states pay a larger growth cost when resources are limiting. Stress also induces a chromosome-number-dependent live-to-dead death hazard that feeds the hypoxia-origin dead compartment. Second, stress can increase chromosome missegregation, while WGD is modeled as a

constant-probability division-associated conversion branch that provides a route into high-ploidy states, generating the karyotypic variation on which selection can act. Third, daughters produced after missegregation pass through a ploidy-dependent survival filter: daughters of lower-ploidy mothers are more vulnerable to lethal dosage imbalance, whereas higher-ploidy mothers can better buffer chromosome gains or losses and convert some CIN events into heritable karyotype variation. These modules map directly onto the Introduction’s mechanistic alternatives by asking whether resource-poor environments suppress high-ploidy proliferation, increase CIN, alter the selective consequences of a constant-probability WGD route, or determine whether altered progeny die or persist.

Observation models. The observation models link each measured data stream to a specific biological component of this mechanism, so the fitted trajectory is constrained by both growth dynamics and chromosome-number evolution:

Data stream	Biological component constrained
Longitudinal tumor burden	Net growth, death, clearance, and resource-dependent proliferation
Terminal single-cell chromosome-number distributions	Karyotype evolution, WGD, missegregation, and survival of altered progeny
Terminal necrotic fraction	Uncleared dead biomass relative to total tumor burden
<i>In vitro</i> passage growth rates	Proliferation under controlled culture and oxygen conditions
<i>In vitro</i> karyotype/chromosome-number and flow-density profiles	Chromosome-number evolution in culture

Joint fitting. We used the same cellular response model for the separate and joint *in vivo*/*in vitro* fits: chromosome-number transitions, hypoxia/resource-stress effects on proliferation and death, stress-induced missegregation, WGD generation, and post-missegregation survival were represented by the same mechanistic structure. The contexts differed in how oxygen trajectories were supplied and in their observation models. In the reported joint analysis, all 14 cross-context mechanistic parameters were soft-coupled. Each received separate *in vivo* and *in vitro* values reconstructed around a transformed-scale center, and a bounded Welsch penalty regularized their separation. The joint fit therefore tested which parameter contrasts were repeatedly recovered across optimizer starts and landscape-informed warm-start pairs. These contrasts are conditional on the model, data streams, regularization, and shared *in vitro* anchor; they identify model-supported hypotheses rather than direct measurements of environmental causes.

Interpretive hierarchy. We distinguished three evidentiary levels throughout the analysis. Experimental burden, chromosome-number, flow-density, and necrosis records are direct observations. Fitted trajectories, latent *in vivo* O_2 , and context-specific response functions are conditional model outputs constrained by those observations. Fixed- O_2 dominant-eigenvector curves are post-fit asymptotic diagnostics under standardized conditions and are therefore more extrapolative than

fitted finite-time trajectories. Numerical optimizer seeds and warm-start pairs assess search dependence; they do not create additional biological replication or sampling-based confidence intervals.

Together, this framework allowed us to ask whether fitted differences between tumors and culture were associated with WGD-favoring or WGD-opposing regimes through resource-dependent proliferation, stress-induced chromosome missegregation, WGD generation, or the ploidy-dependent survival of karyotypically altered progeny.

Results

Lineage-matched culture and tumor trajectories motivate a resource-stress model of ploidy evolution

The motivating observations differed by experimental setting. In culture, longitudinal chromosome-number measurements showed a downward shift after a transient whole-genome-doubling/high-ploidy expansion, demonstrating that the same lineage could first generate and then reduce high-ploidy states (Figure 1B). In tumors, longitudinal burden measurements were paired with chromosome-number and histology/necrosis measurements at harvest; the chromosome-number data therefore compare the starting cell-line references with distinct terminal tumor samples rather than tracking individual tumors longitudinally (Figure 1C). The near-diploid and near-tetraploid SUM159 lineage pairs nevertheless placed the *in vitro* and *in vivo* data on a common interpretive axis (Figure 1A), allowing fitted context contrasts to be compared without treating the two experiments as direct biological replicates.

These observations motivated a model in which resource limitation can both oppose and promote high-ploidy states. In the model overview, resource stress imposes growth and death costs on chromosome-rich states while increasing stress-linked chromosome missegregation; a constant per-division WGD branch provides a separate route into high-ploidy states, and high ploidy can then buffer some missegregation events that would be lethal or strongly deleterious near diploidy (Figure 2). Thus, the predicted ploidy outcome depends on the balance among growth suppression, stress-associated death, CIN generation, the constant-probability WGD route, and post-missegregation survival filtering.

In vitro ploidy evolution under hypoxia occurs without strong direct hypoxia-associated elimination

Separate *in vitro* fits showed that oxygen stress can reshape ploidy without requiring strong direct hypoxia-associated elimination of high-ploidy cells. Passage-growth observations shifted to a lower regime under oxygen deprivation in both starting-ploidy cohorts, and the fitted trajectories reproduced this cohort-level shift while not matching every passage-level observation (Figure 3A). Observed flow-density profiles contained low- and high-DNA-content components under both control and deprivation conditions, with pronounced changes in their relative weights during oxygen deprivation; the fitted model qualitatively reproduced this distributional remodeling across sampled passages (Figure 3B). The corresponding inferred chromosome-state distribution developed a broad high-chromosome mode while retaining a lower-chromosome mode; overlaying the direct single-cell karyotypes and fitted mean trajectory shows where measurements anchor this otherwise

model-inferred passage sequence (Figure 3C). To evaluate whether the shared fitted state also captured lineage-specific mixture weights, we examined the final direct karyotype samples from two parallel oxygen-deprivation lines, O1 and O2, which had been propagated separately from the same 2N precursor through the matched nominal oxygen schedule (Methods; Supplementary Table 3). O1 remained bimodal, with 12 of 20 cells at $N = 46$ –49 and eight at $N = 90$ –100, whereas O2 was dominated by the high-chromosome component, with 16 of 20 cells at $N = 93$ –99. The shared model distribution retained low- and high-chromosome modes but did not reproduce their lineage-specific proportions. The supported result is therefore a shared process capable of generating and expanding high-chromosome states, rather than deterministic prediction of each lineage-specific endpoint.

We next examined whether the fitted remodeling arose through direct elimination of high-ploidy cells or through chromosome-number transitions among their descendants. The post-missegregation survival function increased with chromosome number, allowing a subset of chromosome-loss daughters from high-ploidy mothers to remain viable (Figure 3D–E). Consistent with this mechanism, the paired severe-deprivation trajectories moved in complementary directions: the 2N lineage generated higher-chromosome descendants, whereas the 4N lineage shifted downward, and in both cases direct hypoxia-origin dead burden remained a small component relative to nonviable chromosome-transition products (Figure 3F). In the lowest-objective separate fit, the 4N mean declined from 84.3 to 80.8 chromosomes while direct hypoxia-origin dead burden remained below 1.8% of total burden. Across the ten lowest-objective *in vitro* fits, the corresponding 4N decline was 3.54–4.17 chromosomes and the maximum direct hypoxia-origin dead burden over the daily 4N trajectory remained below 1.9% (Supplementary Table 4). Thus, the fitted result supports bidirectional ploidy redistribution through viable descendants rather than a mechanism dominated by direct hypoxia-associated removal of 4N cells.

In vivo fits capture cohort-scale behavior while retaining multiple compatible parameterizations

The ten lowest-objective separate *in vivo* fits had objectives of 14.119–14.175. Across these fits, log-burden RMSE was 0.663–0.680 across positive observations and 0.628–0.644 after equal weighting of tumor-specific mean squared errors. Terminal mean chromosome-number MAE was 2.41–3.71 chromosomes; terminal-distribution Wasserstein-1 distance was 4.18–5.24 chromosomes and total-variation distance was 0.681–0.714. Reconstructed terminal necrosis-fraction MAE was 0.064–0.103 (Supplementary Table 1). The fits therefore captured broad cohort-level growth and endpoint scales but did not provide uniformly high-fidelity individual-tumor trajectories or terminal distribution shapes. The dynamic parameters were shared within lineage cohorts rather than fitted as mouse-specific latent effects.

The lowest-objective separate fit (seed 25) reproduced the cohort-scale increase in tumor burden while resolving markedly different ploidy trajectories for the two starting states (Figure 4A). The 2N-initiated cohort remained close to diploid throughout tumor growth, whereas the 4N-initiated cohort underwent a pronounced downward shift from near-tetraploid toward a state only slightly above diploid. In both cohorts, this remodeling occurred as fitted effective O_2 declined with increasing burden and modeled dead material accumulated alongside the viable compartment. CIN-associated dead burden constituted the largest modeled dead component by the end of both cohort trajectories, while the relative contribution of direct hypoxia-origin dead burden was greater

in the 4N-initiated cohort. At harvest, the predicted distributions preserved the distinct central ploidy of the observed 2N and 4N cohorts, but were broader than the observed distributions. Thus, the representative fit captured the direction and cohort-level endpoint of ploidy remodeling together with tumor growth, while its broader predicted terminal distributions and model-derived intermediate states preclude interpretation as precise mouse-level trajectory recovery.

Near-equivalent objectives nevertheless arose from substantially different parameter combinations. To determine how this multiplicity mapped onto asymptotic ploidy outcomes, we evaluated all 500 separate-fit parameter sets without refitting at 201 fixed O_2 concentrations from 0% to 5% and classified each dominant eigenvector by whether its dominant mean ploidy was at least 2 or below 2 (Methods; Supplementary Methods). For each O_2 concentration, signed one-feature AUCs summarized both the strength and direction with which each of 18 fitted parameters separated those two outcome classes (Figure 4B). The leading separator changed across the oxygen axis. From 0% through 0.275% O_2 , μ_{hp} was the leading separator; at 0% its AUC magnitude was 0.849, with higher values associated with dominant mean ploidy below 2, whereas λ_{max} had an AUC of 0.761 in the opposite direction. From 0.30% through 0.775% O_2 , γ_μ was the leading separator, again with higher values associated with the lower-ploidy class. The transition region from approximately 0.8% to 0.975% alternated between γ_μ and $p_{mis,base}$. From 1% through 5% O_2 , $p_{mis,base}$ was consistently the leading separator; higher values were associated with dominant mean ploidy at least 2, and its AUC magnitude reached 0.907 at 4.05% O_2 and was 0.903 at 5% O_2 . Stress-linked missegregation and post-missegregation-survival parameters provided additional, more moderate separation over intermediate and high oxygen ranges (Supplementary Table 2). Thus, no single scalar parameter summarized the fitted landscape across the full oxygen range; the dominant descriptive axis shifted from stress-associated death at the lowest oxygen levels to baseline chromosome-missegregation propensity at higher oxygen levels.

To relate this oxygen-dependent separation to the broader fitted landscape, the same parameter rows were aligned with each parameter’s peak AUC and with cluster-wise prior-referenced positions of the 500 separate *in vivo* endpoints (Figure 4B). The *in vivo*-only embedding combined 127,500 retained optimizer-initialization points with those 500 endpoints and used all 18 displayed mechanistic parameters. Clustering the endpoints’ two-dimensional t-SNE positions selected three groups of 43, 412, and 45 solutions (average silhouette 0.655). Among the aligned cluster summaries, κ_O showed the most pronounced separation (descriptive cluster $\eta^2 = 0.401$, rank 1 of 18), with the 45-solution cluster shifted lower than the other two, whereas most parameters showed substantial cross-cluster overlap (Supplementary Table 2). Complete distributions for all 18 parameters are shown in Supplementary Figure 1. These summaries describe parameter regions occupied by compatible fitted solutions; because the cluster construction and displayed parameters came from the same fitted ensemble, they are not independent validation, biological subtypes, or causal “drivers.” Figure 4 therefore separates what the representative fit reproduces (panel A) from what remains non-unique across fitted solutions (panel B). Several parameters also spanned wide ranges among the ten lowest-objective fits, and the fitted burden-error scale reached its upper bound in all ten. The resulting boundary dependence, continuous AUC changes, and multiple fitted-solution clusters are consistent with practical non-identifiability rather than a uniquely resolved biological parameter vector.

Joint fitting yields concordant function-level context contrasts across warm-start-conditioned search ensembles

Joint soft-coupled fits were used to ask which fitted mechanisms remained context-specific when the SUM159 lineage pairs were compared in culture and tumors (Figure 5A). The principal analytical targets were the complete effective missegregation and post-missegregation survival functions; individual scalar parameters were treated as intermediate components of those functions. Six joint warm starts paired objective-minimum representatives from two subclusters within each of three *in vivo* regions of a separate pooled 14-parameter warm-start landscape with the same lowest-objective separate *in vitro* fit (seed 10); each pair was optimized from 500 numerical seeds. To distribute these warm starts, transformed, standardized parameter samples and final solutions from the separate *in vivo* and *in vitro* searches were pooled into that fitted-landscape embedding (Figure 5B; Methods; Supplementary Methods). This pooled warm-start embedding is distinct from the *in vivo*-only 18-parameter landscape in Figure 4B. The marked regions are inputs to joint optimization, not optimized joint-fit endpoints, and the two color scales retain the separate component objectives. All 14 cross-context mechanistic parameters were represented by context-specific values centered on a common transformed-scale value. A bounded Welsch penalty regularized context separation: it is near-quadratic close to zero and saturates for large offsets. Pair-level summaries first aggregated the 500 optimizer starts within each warm-start pair and then compared directions across the six pairs. Optimizer starts quantify numerical search variation and are not biological replicates; moreover, all six pairs share the same *in vitro* anchor. The best retained objective differed across warm-start pairs, ranging from 18.852 to 19.978, so pair-to-pair variation was retained rather than collapsed into a claimed global optimum.

Across the six selected pair winners, the direct in-sample summaries retained the broad burden and terminal mean-chromosome-number scales *in vivo* and the low- versus high-growth regimes *in vitro*, but visible residual scatter and underprediction of several high mean-chromosome-number culture endpoints remained (Figure 5E). These fits therefore provided a basis for comparing fitted functions without establishing uniformly accurate prediction of every observation.

Several parameter directions were concordant across the six sampled warm-start pairs (Figure 5D,F; Supplementary Figure 2). The pair-median *in vivo/in vitro* ratio for the maximal division rate, $\lambda_{\max}$, ranged from 0.099 to 0.223, supporting a lower fitted proliferative ceiling *in vivo*. The maximal stress-induced missegregation increment, p_{misseg} , was higher *in vivo* in all six pairs, with pair-median ratios from 11.1 to 47.4; the *in vivo* value reached its active upper bound in two pairs, so the largest ratios should not be interpreted as precise effect sizes. The oxygen threshold $O_{2,c}$ and post-missegregation viability-loss severity parameter β_{buf} were lower *in vivo* in all six pairs, whereas μ_{hp} , γ_{μ} , $k_{o,\text{mis}}$, and γ_{growth} were higher in all six. Lower β_{buf} increases survival in the fitted function; it is not, by itself, a direct measure of biological buffering strength. Several of these axes were also boundary-limited.

Because the complete effective per-chromosome missegregation probability includes both $p_{\text{mis,base}}$ and the clipped stress-linked increment determined jointly by p_{misseg} , $k_{o,\text{mis}}$, and the death-hazard terms, we evaluated that full function rather than assigning its direction from one parameter (Figure 5D). At 0% O_2 , the six pair-median *in vivo/in vitro* ratios ranged from 11.1 to 48.1 at 44 chromosomes and from 11.2 to 47.8 at 88 chromosomes. At 1% O_2 , the corresponding ranges were 7.43–29.1 and 10.4–42.9. Thus, the complete effective probability was higher *in vivo* at both reference chromosome states in all six pairs under severe and intermediate hypoxia. At 5% O_2 , the

comparison became state dependent: all six pair medians remained above one at 88 chromosomes, but only three did so at 44 chromosomes (Supplementary Table 7).

The three post-misseggregation survival parameters individually showed mixed directions, but their complete nonlinear function gave a consistent and biologically more informative comparison (Figure 5C). Pair-median per-copy survival ranged from 0.728 to 0.952 at 44 chromosomes and from 0.901 to 0.958 at 88 chromosomes *in vivo*, compared with 0.204 and 0.837, respectively, *in vitro*. Absolute survival was therefore higher *in vivo* at both reference states in all six pair medians and all 3,000 numerical solutions. Conversely, the fitted 44-to-88 chromosome survival gradient was 0.006–0.186 *in vivo* versus 0.633 *in vitro*, and was larger *in vitro* in every pair and numerical solution (Supplementary Table 7). The supported result is therefore two-part: altered daughters have higher absolute fitted survival in tumors, while culture has the stronger fitted ploidy-dependent survival gradient. The 5th–95th percentile intervals across the 500 numerical seeds were narrow within most pairs, but these intervals describe optimizer-start spread, not biological confidence intervals.

The bounded Welsch penalty limits how strongly large context separations are regularized. Among the ten lowest-objective solutions from each pair, the summed soft-coupling penalty occupied 75.8–89.5% of its 14-parameter maximum, and several individual terms were nearly saturated. All 3,000 solutions had at least one active parameter bound, all differential-evolution runs stopped under the configured relative-tolerance/step-tolerance rule before the 500-iteration target, and all attempted local refinements were rejected (Supplementary Figure 3). The observed directional concordance is therefore limited to the six sampled warm-start-conditioned search ensembles; it does not resolve the global optimum or behavior in unsampled regions of the objective landscape and does not identify unique context effect sizes.

Fixed-O₂ analyses expose an uncertainty architecture rather than a unique response regime

The separate *in vivo* calibration left multiple parameterizations that could explain the observed ploidy endpoints. We characterized this uncertainty with fixed-O₂ dominant-eigenvector response curves: for each candidate fit, the analytical attractor was evaluated across an oxygen grid and grouped by the regression-smoothed shape of the predicted O₂–ploidy relationship (Methods; Supplementary Methods). Among the 500 separate *in vivo* fits, 316 were complex nonmonotone, 50 inverted-U-shaped, 50 monotone increasing, 47 U-shaped, 26 approximately flat, five increase-then-plateau, four decrease-then-plateau, and two monotone decreasing (Figure 6A). Spectral-gap diagnostics classified only 141 fits as reliable, 168 for caution, and 191 as unreliable; among the selected separate-*in vivo* warm-start representatives, three were caution and three were unreliable.

Because these curves are asymptotic operator diagnostics rather than fitted finite-time observations, small spectral gaps weaken their biological interpretation over experimental time horizons. We next evaluated how the six selected joint-fit winners organized ploidy responses across a standardized oxygen–misseggregation input plane (Figure 6B). Holding the remaining fitted parameters of each winner fixed, complete 201 × 60 response surfaces were calculated across oxygen and a forced effective per-chromosome misseggregation probability. The surfaces did not collapse to one common response topology: the extent of low- and high-ploidy regions and the weak-spectral-gap boundaries differed across the six warm-start-conditioned winners, and their model-implied fitted trajectories traversed different parts of those surfaces. Thus, the directional joint-fit function contrasts did not imply a uniquely resolved global oxygen–CIN–ploidy landscape. These standardized surfaces are

post-fit model diagnostics, not independent CIN measurements or causal oxygen-by-CIN interventions.

Finally, response class and the complete fitted *in vivo* MAP objective were overlaid on the pooled 14-parameter t-SNE coordinate system used for joint warm-start selection. Several common response classes overlapped in parameter space and objective value, and no single response class uniquely contained the lowest-objective solutions. This diagnostic therefore retained competing regimes rather than selecting one response shape by assumption (Figure 6C).

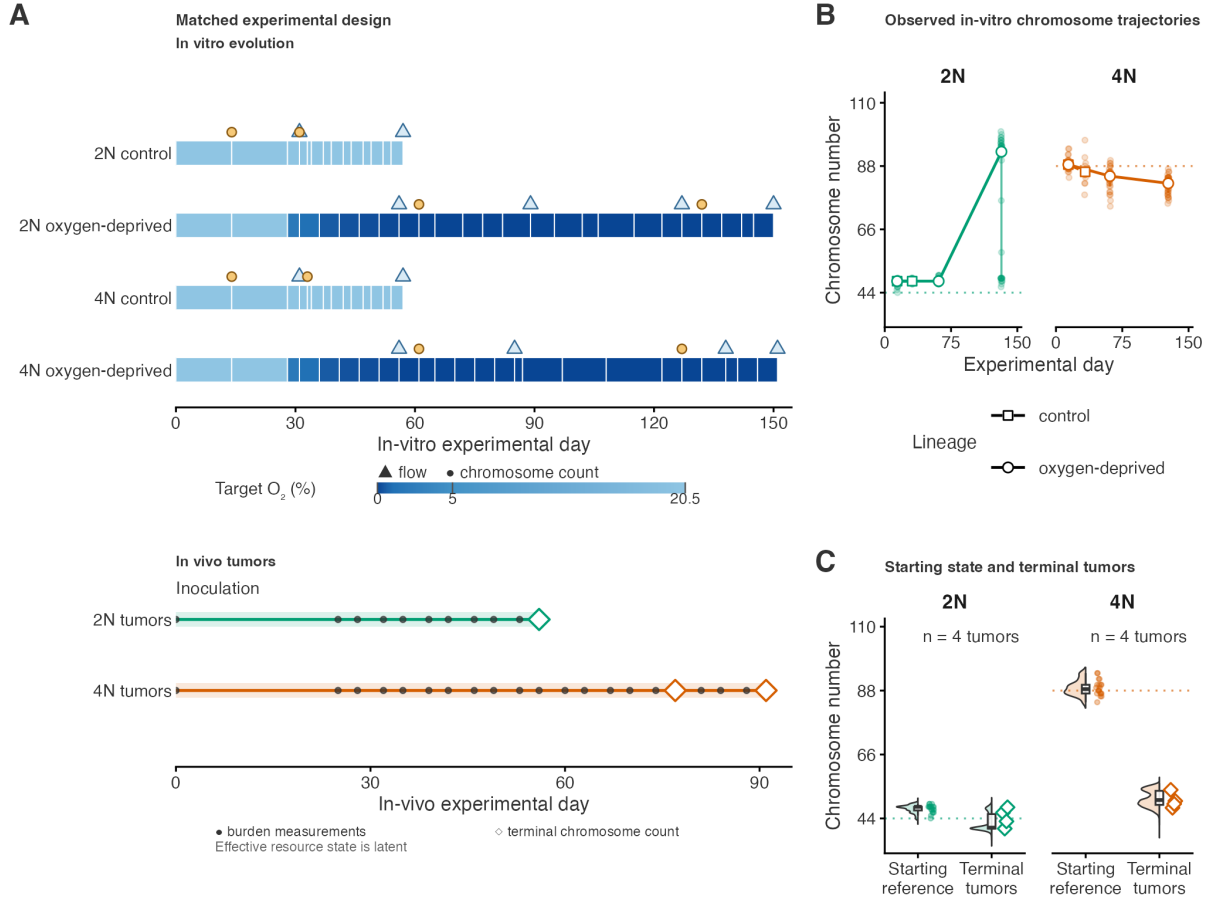


Figure 1: Matched experimental systems and motivating chromosome-number changes. (A) Simplified design for matched SUM159 near-diploid (2N) and near-tetraploid (4N) lineages. Culture strips show the longest tracked control and oxygen-deprived paths, with known target O_2 encoded from light to dark; triangles mark flow cytometry and circles mark chromosome counts. Tumor strips show burden-measurement times and terminal chromosome sampling. Tumor resource state was not measured and is latent in the model. (B) Observed cell-level chromosome numbers in culture. Lines and ribbons show medians and interquartile ranges; the shared starting reference begins both summary trajectories. Control series end earlier than oxygen-deprived series. (C) Starting cell-line reference karyotypes and terminal tumor-cell chromosome distributions. Boxes show distribution summaries, violins show terminal cell distributions, and diamonds show individual tumor medians (four tumors per starting-ploidy cohort). Starting and terminal measurements are distinct samples rather than longitudinal karyotyping of the same tumor.

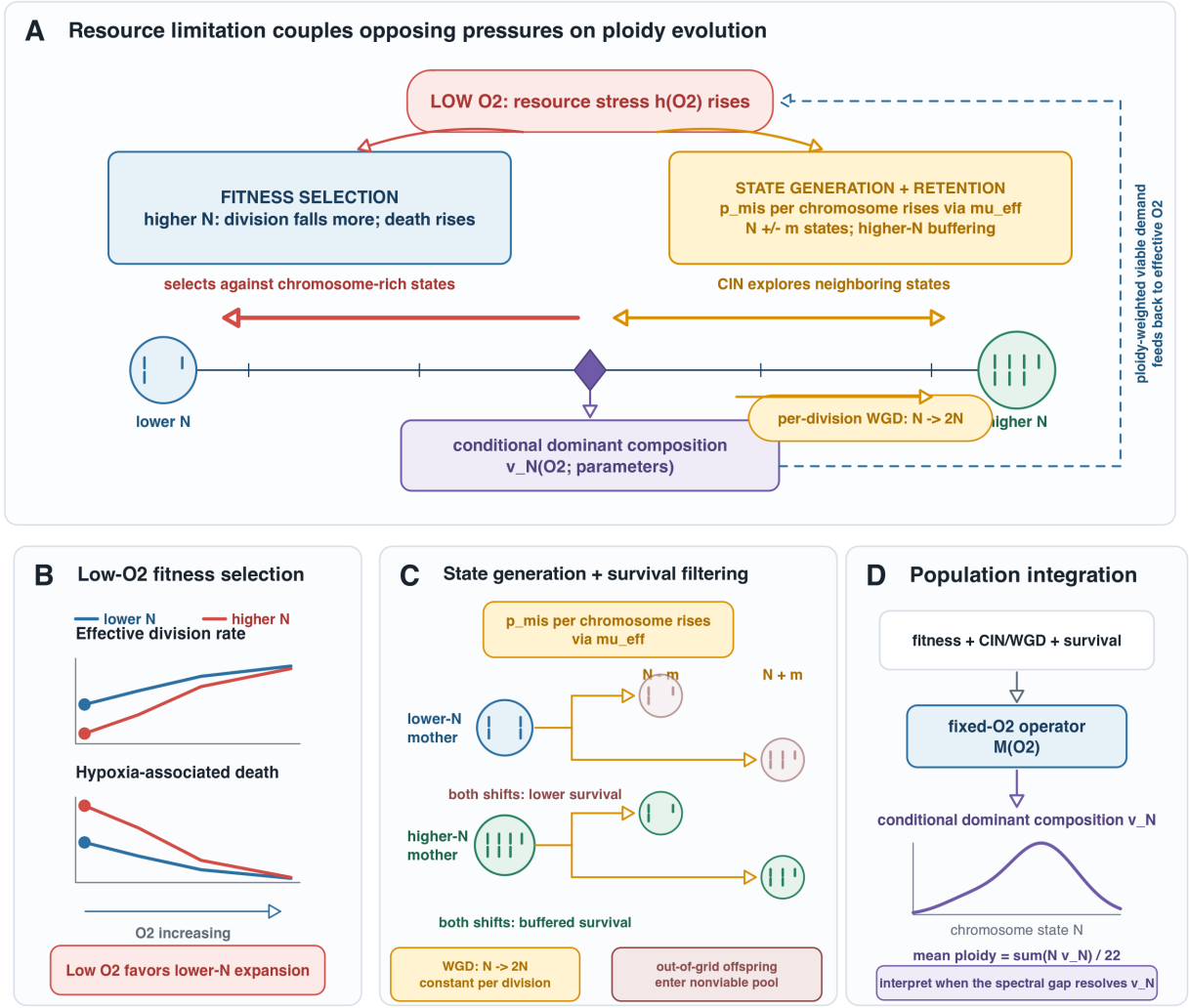


Figure 2: Model overview: resource limitation can impose opposing pressures on ploidy evolution. (A) Low oxygen raises bounded resource stress and simultaneously strengthens selection against chromosome-rich states while increasing the per-chromosome missegregation probability that supplies chromosome-state variation. In the dynamic model, viable burden and ploidy-weighted oxygen demand close the feedback loop; the resulting composition depends jointly on oxygen and the fitted model parameters. (B) Under the ploidy-related stress mode, low oxygen suppresses effective division and increases hypoxia-associated death more strongly in higher-chromosome-count states, creating a relative expansion advantage for lower- N states. (C) The per-chromosome missegregation probability rises through the stress-linked death hazard and generates symmetric $N - m$ and $N + m$ daughter branches. For a given $|m|$, altered daughters from higher- N mothers have greater post-missegregation survival. Constant per-division WGD provides a separate tracked-lineage transition $N \rightarrow 2N$, and out-of-grid offspring enter the CIN-associated nonviable pool. (D) Growth, death, CIN/WGD transitions, and post-missegregation survival define the fixed- O_2 population operator. When its dominant mode is sufficiently separated, the normalized dominant eigenvector gives the conditional asymptotic chromosome-state composition and mean ploidy.

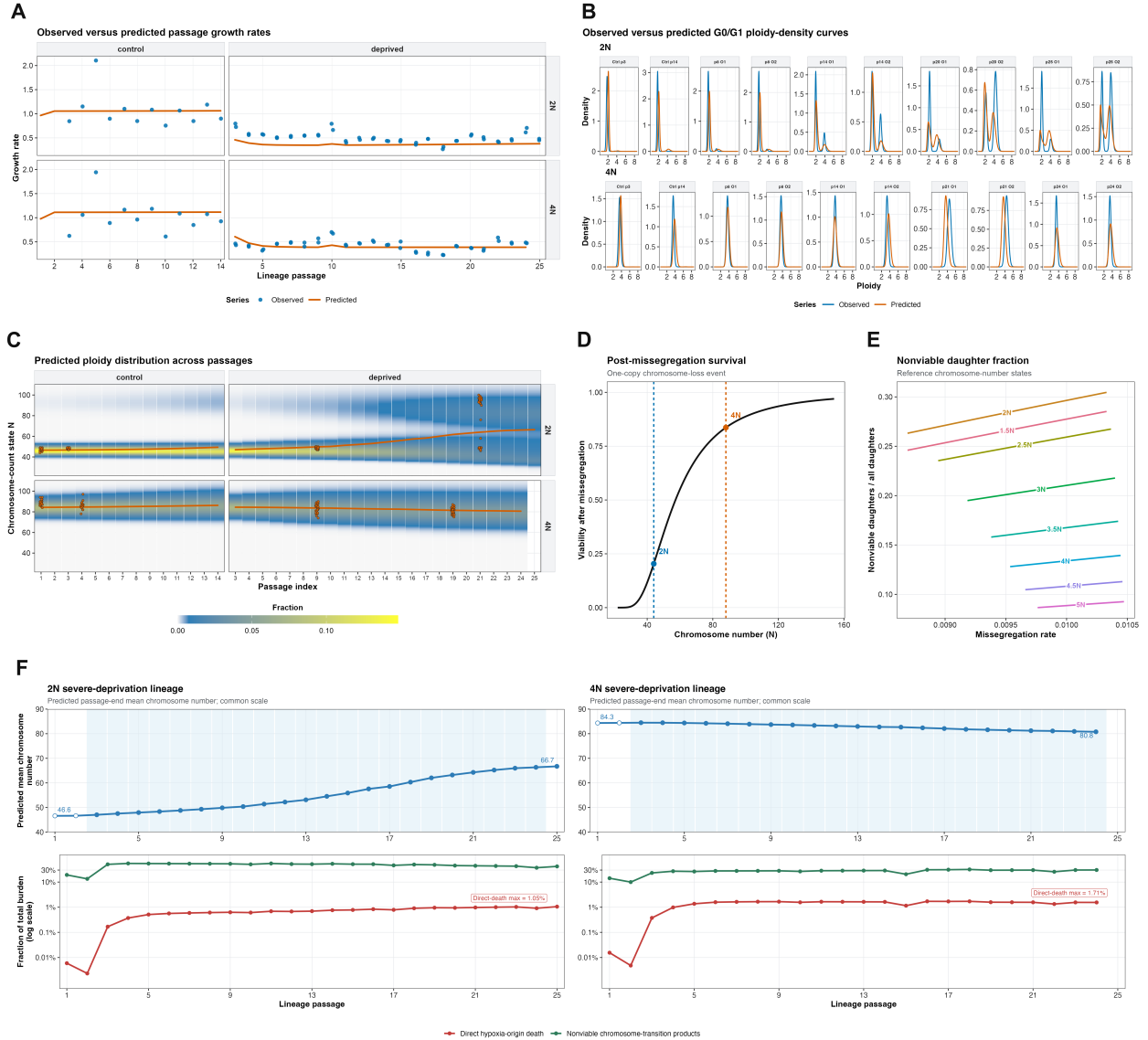


Figure 3: In vitro ploidy reshaping occurs without strong direct hypoxia-associated elimination. (A) Observed (blue points) and fitted (orange line) passage growth rates from the lowest-objective retained separate *in vitro* fit (seed 10), arranged as 2N/4N rows and control/deprived columns. The fitted trajectories reproduce the lower growth-rate regime under oxygen deprivation while individual passage measurements remain variable. (B) Observed (blue) and fitted (orange) G0/G1 DNA-ploidy density curves from the same fit. Parallel O1 and O2 measurements at the same deprived passage are displayed in separate sample-level subpanels rather than connected or overlaid as one observed curve; the corresponding shared prediction is repeated in each subpanel. Both sources are drawn as thin solid lines, with the fitted curve layered above the observed curve where they overlap. High-DNA-content components are present under both control and deprivation conditions, while their relative peak weights change prominently during deprivation; the shared fitted state captures the qualitative remodeling but not every lineage-specific peak proportion or width. (C) The predicted chromosome-state distribution across passages shows the oxygen-deprived 2N lineage developing a high-chromosome component while retaining a lower-chromosome mode. The orange line gives the fitted mean chromosome number and the orange points with dark outlines give direct single-cell karyotypes; these measurements occur only at sampled passages, whereas the intervening heatmap is model inferred. (D) The fitted post-misseggregation survival function increases with chromosome number. The black curve is evaluated at the diploid $N = 44$ and tetraploid $N = 88$ reference states, highlighted by the blue 2N and orange 4N dashed lines and their same-color curve intersections. (E) The model-implied nonviable daughter fraction plotted against missegregation rate shows how chromosome-loss opportunity and

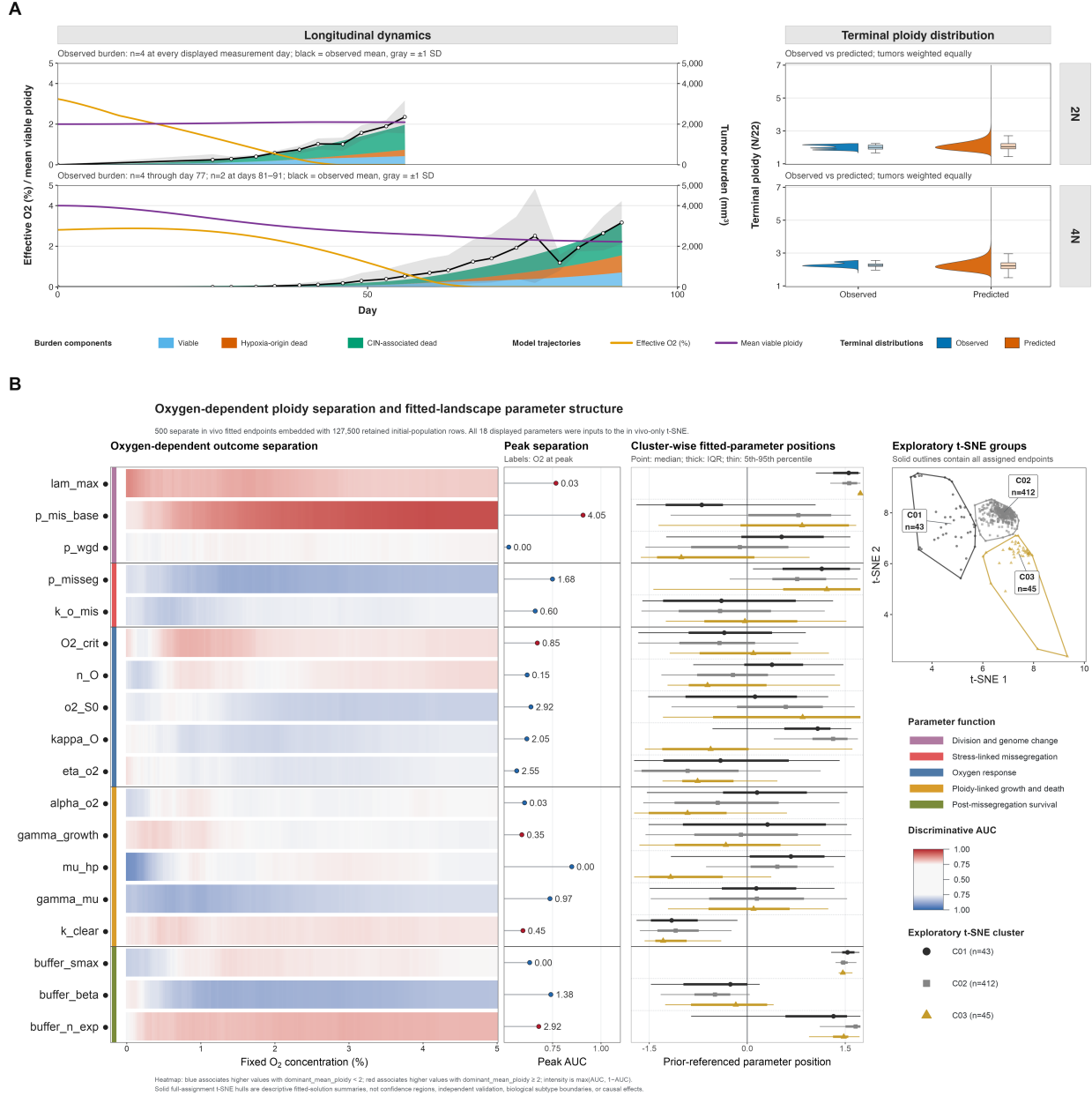
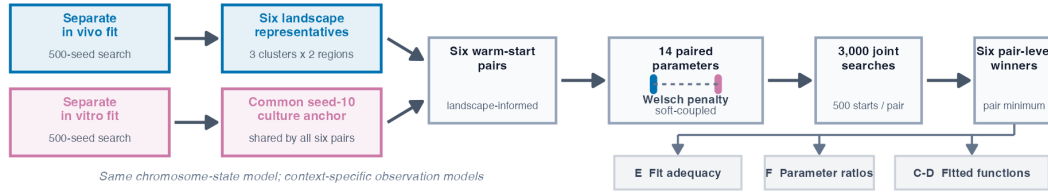


Figure 4: Separate in vivo fits capture burden, resource, and ploidy dynamics while retaining multiple resource-response parameterizations. (A) Cohort-level summary of the lowest-objective retained separate fit (seed 25). The left column shows 2N-initiated and 4N-initiated longitudinal dynamics on a common 0–100-day axis. Stacked areas, averaged over the tumor trajectories available at each day, are read from the right axis and show predicted viable, direct hypoxia-origin dead, and CIN-associated dead tumor burden. The black points and connecting curve show observed cohort-mean burden at the actual measurement days; gray bands show one sample standard deviation. The 2N summaries contain four tumors throughout. The 4N summaries contain four tumors through day 77 and two tumors at days 81–91. Orange and purple curves show cohort-mean fitted effective O₂ and viable-cell mean ploidy, respectively, on the shared numerical 0–5 left axis. Mean ploidy is viable-cell weighted mean chromosome number divided by $N_{\text{unit}} = 22$; sharing this numerical axis does not equate the physical units of O₂ and ploidy. The right column reconstructs the report’s terminal observed-versus-predicted chromosome-number comparison after conversion to ploidy, using blue for observed and orange for predicted distributions. In each half-violin/box summary, the left half shows the weighted distributional shape and the box-and-whisker at right shows the weighted median and quartiles. Each tumor receives equal total weight within its cohort and source, so tumors with more profiled cells do not dominate the observed distribution. Effective O₂, burden compartments, and intervening ploidy paths are model predictions;

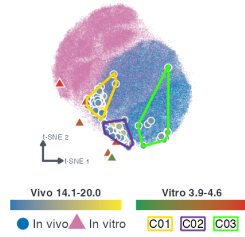
Joint-fit design, adequacy, and context-specific fitted functions

Six landscape-informed joint-pair winners. Direct fits are in-sample; across-winner ranges and traces show solution sensitivity, not confidence intervals.

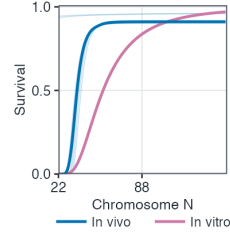
A Joint-fitting workflow



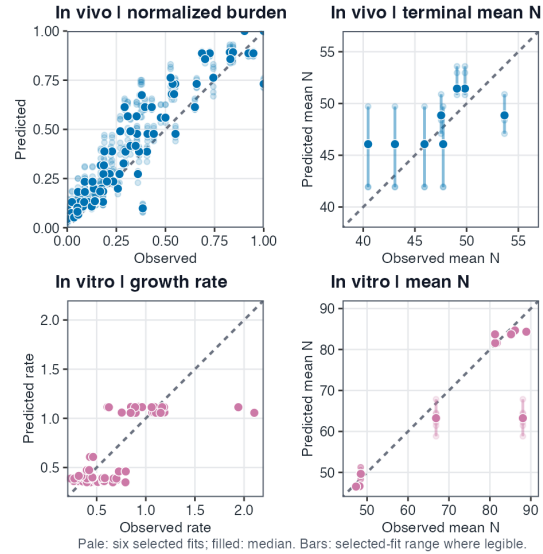
B Warm-start map



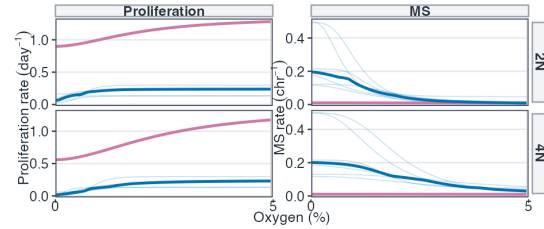
C Survival



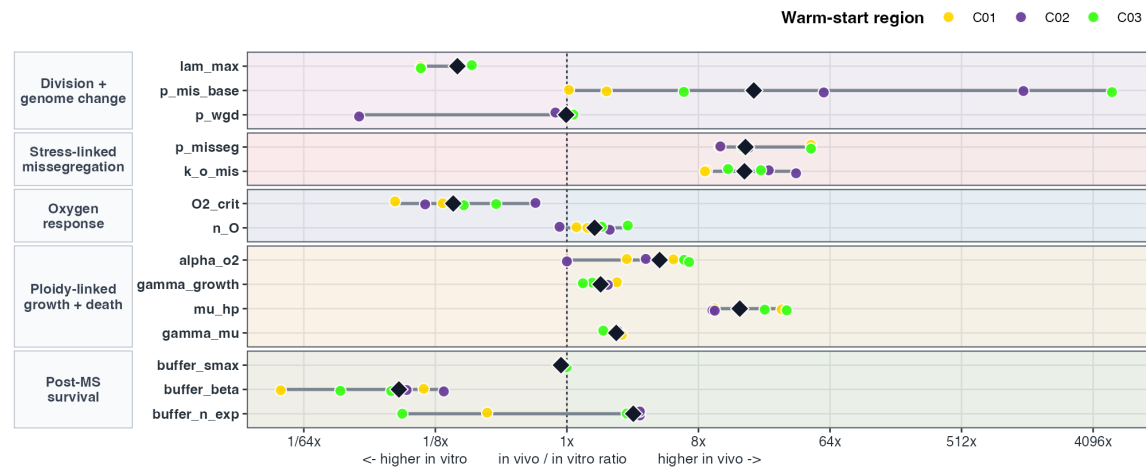
E In-sample fit adequacy



D Oxygen-dependent fitted functions



F In-vivo / In-vitro parameter ratios



Six values/parameter; gray = range; black diamond = median.

Common log2 ratio scale for all 14 coupled parameters.

C-D: thin curves = six fits; thick = pointwise median. Necrosis omitted: predictions unavailable. B-F use the same six winners; D is display-interpolated to a declared 0-5% O2 grid.

Figure 5: Joint-fit design, in-sample adequacy, and context-specific fitted functions.

(A) Joint-fitting workflow. Separate 500-seed *in vivo* and *in vitro* searches form two input lanes.

Six landscape-informed *in vivo* representatives—two from each of three exploratory regions—are paired with the common seed-10 *in vitro* anchor. Each of the six pairs enters the same chromosome-state model with 14 paired context-specific parameters linked by the Welsch soft-coupling penalty.

Five hundred numerical starts per pair give 3,000 joint searches; the lowest retained objective within each pair supplies the six winners summarized in panels C–F.

(B) Pooled separate-fit t-SNE landscape used to distribute joint warm starts. Initial samples are blue circles for *in vivo* and

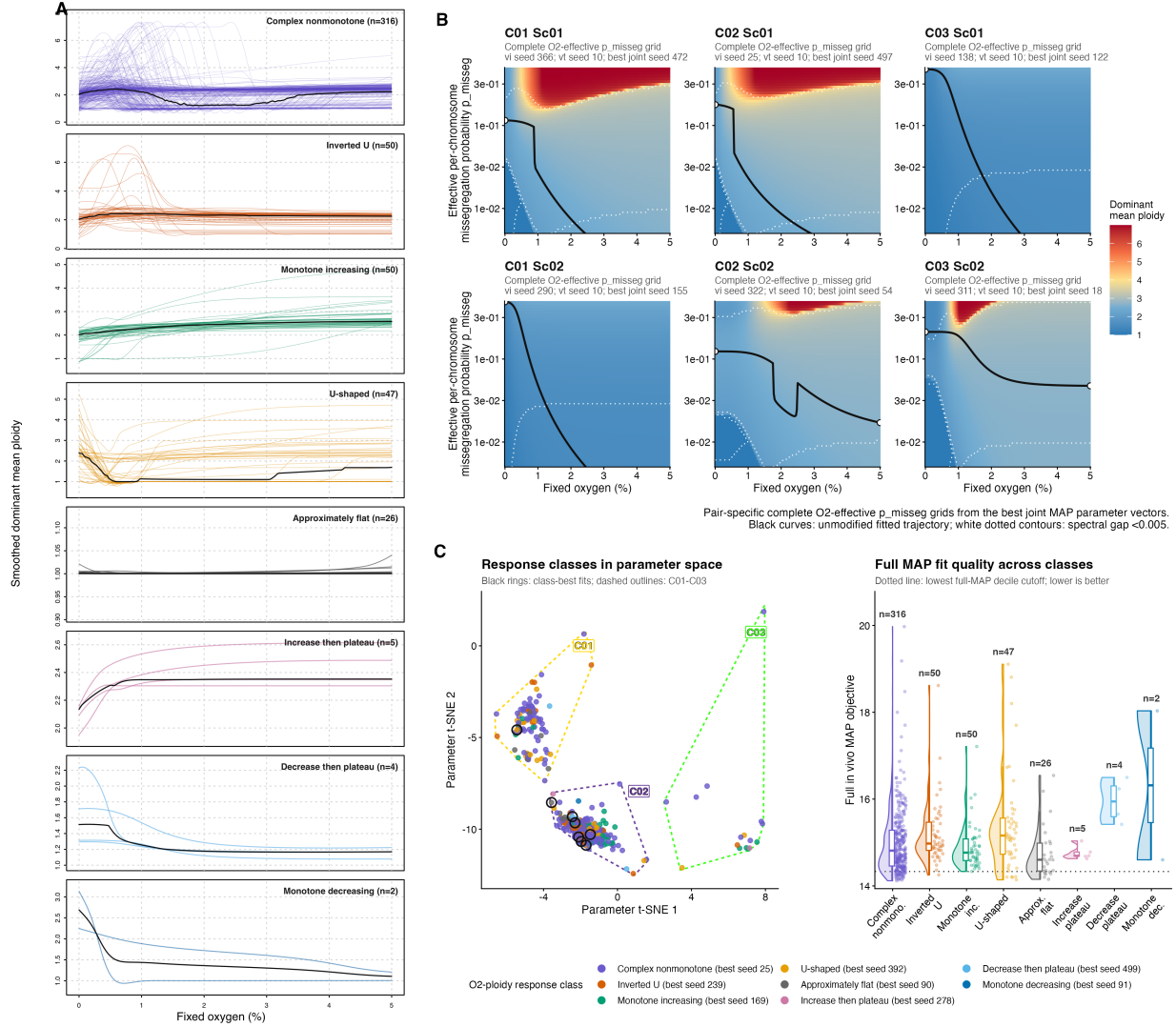


Figure 6: Fixed-O₂ diagnostics retain competing asymptotic oxygen-CIN-ploidy responses. (A) Analytical dominant-eigenvector ploidy curves from 500 separate *in vivo* fitted solutions were grouped by regression-smoothed response shape. The eight response classes are arranged in one column with shared axes; class names and sizes are printed within the upper-right corner of each panel. Individual solutions are colored, and the black curve is the pointwise class median. These pooled class counts include reliable, caution, and unreliable spectral-gap strata. (B) Pair-specific post-fit response surfaces for the six selected joint winners, arranged with Sc01/Sc02 as rows and C01/C02/C03 as columns. Each response-surface plotting region is square, and its title uses the corresponding C01 orange, C02 purple, or C03 green warm-start-region color. For each winner, dominant mean ploidy was evaluated on a complete grid of 201 oxygen concentrations and 60 forced effective per-chromosome missegregation probabilities while all remaining fitted parameters were held fixed. Black curves show the model-implied population-average effective missegregation probability along that winner's unmodified fixed-O₂ trajectory; white dotted contours mark spectral gap < 0.005, where the leading operator mode is weakly separated. The surface is a standardized model-response diagnostic, not an independently fitted CIN measurement or a causal oxygen-by-CIN intervention. (C) Left, the 500 separate *in vivo* fitted solutions are positioned in the pooled 14-parameter t-SNE coordinate system used for the joint warm-start landscape in Figure 5B and colored by fixed-O₂ response class. This is distinct from the *in vivo*-only 18-parameter embedding in Figure 4B. The eight black rings mark the single lowest-objective fit within each response class, and the legend gives the corresponding best-seed number in parentheses. Dashed outlines delimit the C01, C02, and C03 warm-start regions using the same orange, purple, and green colors as Figure 5. Right, complete MAP objective distributions are shown with response class on the horizontal axis and full MAP objective on the vertical axis. For each class, the left

Discussion

Here, we integrated lineage-matched *in vitro* and *in vivo* ploidy-evolution datasets into a common mechanistic framework. The main result is not a uniquely identified parameter vector, but a small set of function-level contrasts that remained directional across six warm-start-conditioned search ensembles. Within the fitted model, tumors had a lower proliferative ceiling and higher complete effective per-chromosome missegregation probabilities at 0% and 1% O₂. At the same time, altered daughters had higher absolute fitted per-copy survival *in vivo* at both 44 and 88 chromosomes, whereas culture had the stronger fitted survival increase between those states. Absolute altered-daughter survival and its ploidy-dependent gradient are therefore separable axes: a context can retain altered daughters more readily overall without imposing the stronger ploidy dependence on that retention.

This separation sharpens the biological significance of the model. Resource context may change not only how much chromosome-number variation is generated, but also how chromosome state filters whether that variation persists. In that sense, the model links CIN production to the retention of viable karyotypic diversity rather than treating CIN, WGD, and survival as interchangeable proxies. The complete-function comparison was essential: directions of individual survival parameters were mixed, whereas the evaluated nonlinear survival function yielded the two-part context contrast in all six pair medians and all 3,000 numerical solutions.

These results remain conditional. Soft coupling permits context-specific values while regularizing their separation on the transformed parameter scale, but the reported Welsch penalty is bounded, so sufficiently large differences contribute a saturated penalty rather than an indefinitely increasing cost. Repeated separation across warm-start pairs therefore identifies model-supported hypotheses conditional on the likelihoods, bounds, regularization, and common *in vitro* anchor; it is not an equality test and does not establish that environmental context caused the fitted difference. Likewise, fitted effective O₂ and stress-linked parameter contrasts are latent model quantities, not direct measurements of tissue oxygen, CIN, or death mechanisms.

The explicit *in vitro* model limitation is also informative. Observed flow-density and chromosome-count distributions support the presence and remodeling of low- and high-ploidy components, while the late O1 and O2 samples differ in the relative occupancy of those components. The model generated a shared mixed distribution but did not reproduce the lineage-specific late composition of both lines. This localizes the limit of the shared modeled process: lineage-specific latent histories, stochastic diversification, measurement heterogeneity, or other unmodeled processes would be needed to explain the divergent compositions. The supported claim is therefore a common process capable of generating and filtering a mixed distribution, rather than complete reconstruction of lineage-specific evolution.

Several observations limit a purely resource-dependent interpretation of ploidy evolution. First, the present model does not include a separate oxygen-independent cost of abnormal chromosome content. Aneuploid states can incur proteotoxic, replication, mitotic, and gene-dosage stresses even outside the specific oxygen/resource term represented here.^{7–9} Consequently, the fitted resource-dependent penalty should not be interpreted as an exhaustive account of high-ploidy or aneuploid fitness costs.

Second, oxygen is only one component of the broader resource environment experienced by tumor cells. The analyzed datasets did not directly measure nutrient availability, perfusion, local

consumption, or spatial oxygen heterogeneity. The fitted *in vivo* O₂ trajectory is therefore a latent model variable, and nominal *in vitro* oxygen and fitted *in vivo* oxygen should not be treated as experimentally equivalent energetic states. Oxygen is informative within the model but remains an incomplete proxy for resource stress.

Third, the model omits other mechanisms that could alter the generation or filtering of high-ploidy states, including context-dependent WGD generation, tissue-specific polyploidization programs, and immune or stromal selection. The present SUM159 analysis cannot distinguish these unmodeled possibilities from resource-mediated effects. Its conclusions should therefore remain restricted to the fitted mechanisms and matched culture/tumor datasets rather than generalized to all tissue-specific ploidy differences.

The fitted contrasts remain limited by practical identifiability and search coverage. Near-equivalent separate *in vivo* objectives were obtained from boundary-dependent parameter combinations, and every joint solution had at least one active bound. The six joint pairs sampled six exploratory *in vivo* subclusters but only one *in vitro* warm start; their best objectives differed by more than the within-pair numerical spread. The bounded Welsch penalty was close to saturation for many large context splits, so it did not continue to impose strong shrinkage once those splits became large. Fixed-O₂ interpretation was also limited by small spectral gaps: only 28.2% of separate fits were classified as reliable, and none of the six selected separate-*in vivo* warm-start representatives met that criterion. The directional concordance reported here is limited to these six warm-start-conditioned search ensembles; a unique effect size, global optimum, and behavior in unsampled objective-landscape regions remain unresolved.

Two implementation limitations require separate caution. First, out-of-range chromosome-transition mass is routed to the CIN-associated dead compartment, so that compartment combines biological nonviability with finite-grid boundary loss. Second, the necrosis likelihood is active in the objective, but the current per-sample `necrosis_fit.tsv` exports contain missing predicted values; the terminal necrosis-fraction audit was therefore reconstructed from the stored model states and objective components. The export should be repaired before claiming fully automated row-level reproduction of the necrosis fit.

These findings also suggest hypotheses for therapies that induce chromosome-segregation errors. Many anticancer agents elevate chromosome mis-segregation either directly, by disrupting mitosis, or indirectly, by producing replication and DNA-damage lesions that persist into mitosis.^{17–23} Within the fitted ploidy-dependent survival framework, such agents are predicted not to have uniform effects across chromosome-number states: near-diploid cells are predicted to be more vulnerable to missegregation-induced dosage imbalance, whereas WGD or high-ploidy cells may better buffer individual chromosome gains or losses and convert some errors into heritable karyotype variation. This offers a model-based hypothesis for prior drug-response analyses in which multiple mitotic spindle poisons, including paclitaxel, vinblastine, and vinorelbine, were among the strongest ploidy-associated agents.²⁴ It also suggests a possible explanation for why gemcitabine, despite being classically viewed as an S-phase antimetabolite, may show a strong association with ploidy: independent chromosome-loss assays have identified gemcitabine as a potent inducer of chromosome instability.¹⁷ Thus, replication-stress-induced chromosome mis-segregation is a candidate ploidy-dependent therapeutic vulnerability generated by the model, not a response tested in the present SUM159 experiments.

1 Supplementary Information

1.1 *In vivo* experiments

The *in vivo* analysis used near-diploid and near-tetraploid variants of the SUM159 triple-negative breast cancer cell line, denoted SUM159_NLS_2N_A6M and SUM159_NLS_4N_A4M, respectively. For each tumor, 1×10^6 cells were injected orthotopically into mice. The subset analyzed here comprised eight tumors: four initiated from the near-diploid 2N line and four initiated from the near-tetraploid 4N line.

- **2N cohort:** four mice injected with SUM159_NLS_2N_A6M.
- **4N cohort:** four mice injected with SUM159_NLS_4N_A4M.

Tumors were harvested at predefined endpoints, and single-cell RNA sequencing was performed on all eight tumors. Chromosome-number distributions used for model calibration were derived from these endpoint single-cell profiles and analyzed together with the corresponding tumor-burden trajectories. Only this untreated subset was used for the modeling analyses presented here. Processed data and associated experimental metadata were organized and retrieved through CLONEID, a framework for structuring clone-specific genotypic and phenotypic measurements.²⁵

Supplementary Table 1: Absolute fit-quality metrics for the ten lowest-objective separate *in vivo* fits. Burden errors are calculated on the log-burden scale. Terminal chromosome-number metrics and reconstructed necrosis-fraction MAE are averaged equally across tumors/harvests.

Rank	Fit	Objective	Burden log-RMSE	Tumor-balanced log-RMSE	Terminal mean- <i>N</i> MAE	Terminal Wasserstein-1	Terminal total variation	Reconstructed necrosis-fraction MAE
1	seed25	14.1193	0.6660	0.6317	2.41	4.51	0.697	0.091
2	seed366	14.1340	0.6680	0.6341	2.54	4.75	0.708	0.078
3	seed292	14.1372	0.6724	0.6384	2.79	4.59	0.698	0.064
4	seed392	14.1406	0.6784	0.6426	3.33	4.92	0.689	0.093
5	seed90	14.1524	0.6799	0.6439	3.65	4.86	0.684	0.092
6	seed391	14.1553	0.6797	0.6439	3.38	4.75	0.682	0.091
7	seed264	14.1558	0.6735	0.6400	2.56	4.18	0.682	0.079
8	seed109	14.1724	0.6629	0.6284	2.85	4.30	0.681	0.103
9	seed322	14.1724	0.6677	0.6331	2.52	4.86	0.714	0.078
10	seed26	14.1748	0.6784	0.6425	3.71	5.24	0.698	0.094

Note: Positive-observation burden RMSE weights every retained measurement equally; tumor-balanced RMSE first averages squared residuals within each tumor and then weights tumors equally. Wasserstein-1 and mean-*N* MAE are reported in chromosomes. Necrosis predictions were reconstructed from matched dead-total and total predicted burden volumes because the exported row-level predicted-necrosis column was missing. The ten optimizer-selected fits are not biological replicates or confidence-interval samples.

1.2 *In vitro* experiments

The *in vitro* analysis used matched near-diploid and near-tetraploid SUM159 lineages. For each starting ploidy, a control branch was maintained at 20.5% O₂, and two parallel oxygen-deprivation lineages (O1 and O2) were stepped over successive passages through target levels of 2%, 1%, 0.5%, 0.3%, 0.2%, 0.1%, and 0% O₂. The main deprivation branch was then maintained at target 0% O₂; an additional branch after the first 0% passage was evaluated at target 1% O₂. We describe O1 and O2 as parallel lineages because the available records do not establish whether they should be classified as biological replicates.

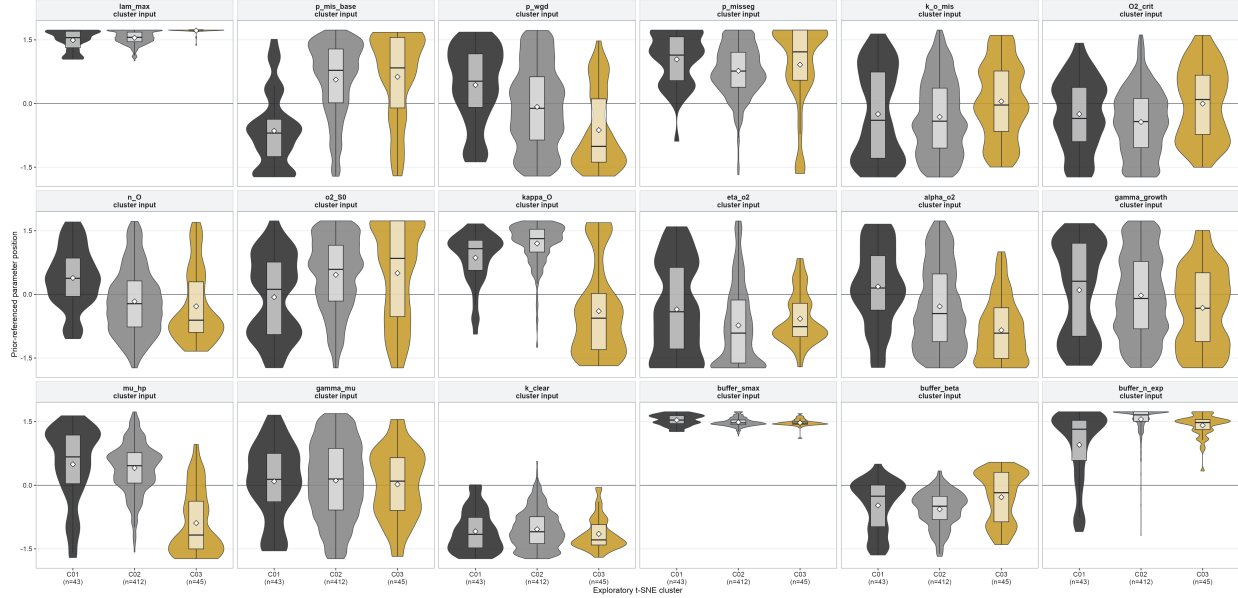
Supplementary Table 2: Continuous fixed-O₂ single-parameter separation underlying Figure 4B. Section A gives contiguous oxygen intervals over which the same parameter was the leading discriminator of dominant mean ploidy ≥ 2 versus < 2 . Section B gives exact landmark values cited in the Results. Section C reports the exploratory-cluster counts and diagnostics used to interpret the aligned fitted-landscape component.

Section	O ₂ (%)	Parameter	Direction of higher values	Raw AUC	AUC minimum	AUC maximum	Peak O ₂ (%)
A. Contiguous intervals of the leading single-parameter discriminator							
A	0–0.275	mu_hp	Higher values: $\bar{P}_{\text{dom}} < 2$	–	0.702	0.849	0
A	0.3–0.775	gamma_mu	Higher values: $\bar{P}_{\text{dom}} < 2$	–	0.696	0.736	0.625
A	0.8–0.825	p_mis_base	Higher values: $\bar{P}_{\text{dom}} \geq 2$	–	0.712	0.715	0.825
A	0.85–0.875	gamma_mu	Higher values: $\bar{P}_{\text{dom}} < 2$	–	0.715	0.719	0.875
A	0.9–0.95	p_mis_base	Higher values: $\bar{P}_{\text{dom}} \geq 2$	–	0.732	0.735	0.95
A	0.975	gamma_mu	Higher values: $\bar{P}_{\text{dom}} < 2$	–	0.737	0.737	0.975
A	1–5	p_mis_base	Higher values: $\bar{P}_{\text{dom}} \geq 2$	–	0.740	0.907	4.05
B. Exact AUC values at Results landmarks							
B	0	mu_hp	Higher values: $\bar{P}_{\text{dom}} < 2$	0.151	0.849	0.849	–
B	0	lam_max	Higher values: $\bar{P}_{\text{dom}} \geq 2$	0.761	0.761	0.761	–
B	1	p_mis_base	Higher values: $\bar{P}_{\text{dom}} \geq 2$	0.741	0.741	0.741	–
B	1	gamma_mu	Higher values: $\bar{P}_{\text{dom}} < 2$	0.267	0.733	0.733	–
B	1	buffer_beta	Higher values: $\bar{P}_{\text{dom}} < 2$	0.276	0.724	0.724	–
B	1	p_misseg	Higher values: $\bar{P}_{\text{dom}} < 2$	0.293	0.707	0.707	–
B	4.05	p_mis_base	Higher values: $\bar{P}_{\text{dom}} \geq 2$	0.907	0.907	0.907	–
B	5	p_mis_base	Higher values: $\bar{P}_{\text{dom}} \geq 2$	0.903	0.903	0.903	–
B	5	p_misseg	Higher values: $\bar{P}_{\text{dom}} < 2$	0.311	0.689	0.689	–
B	5	buffer_beta	Higher values: $\bar{P}_{\text{dom}} < 2$	0.328	0.672	0.672	–
B	5	buffer_n_exp	Higher values: $\bar{P}_{\text{dom}} \geq 2$	0.655	0.655	0.655	–
C. Exploratory t-SNE cluster structure and displayed-parameter audit							
Section	Cluster or item	Count	Selected k	Average silhouette	η^2_{cluster}	Rank	Maximum pairwise mean difference
C	C01	99	3	0.816	–	–	–
C	C02	385	3	0.816	–	–	–
C	C03	16	3	0.816	–	–	–
C	Displayed parameters	18	–	–	–	–	–
C	Cluster-construction inputs	14	–	–	–	–	–
C	Projected-only parameters	4	–	–	–	–	–
C	p_mis_base (largest separation)	–	–	–	0.613	1	2.047

Note: The positive class is dominant mean ploidy ≥ 2 . Discriminative AUC is $\max(\text{AUC}, 1 - \text{AUC})$; the direction column retains the sign of the raw AUC. Interval rows summarize all 201 oxygen values, whereas landmark rows provide exact values cited in the text. Cluster η^2 is between-cluster sum of squares divided by total sum of squares on the prior-referenced scale. The 500 fitted endpoints are optimizer-derived solutions, not biological replicates; the cluster quantities are descriptive rather than independent tests.

All 18 fitted parameters across exploratory *in vivo* t-SNE clusters

Prior-referenced positions: all 18 parameters were inputs to the *in vivo*-only t-SNE and best-endpoint cluster construction.



Supplementary Figure 1: **All 18 fitted parameters across the exploratory separate-*in vivo* t-SNE clusters.** The 18 parameter facets are arranged in three rows and six columns. Each facet shows the prior-referenced parameter-position distribution for clusters C01, C02, and C03; boxplots show medians and interquartile ranges and diamonds show means. “Cluster input” marks that every displayed parameter contributed to the *in vivo*-only 18-parameter t-SNE and best-endpoint cluster construction. Zero is the midpoint of the configured transformed bound interval and one unit is the standard deviation of a uniform distribution over that interval. These distributions describe optimizer-derived fitted endpoints and do not constitute independent validation or biological replication.

For calibration, each fit entry retained the assigned oxygen level, passage duration, initial and final corrected cell counts, and a segment-level growth-rate estimate. Chromosome-number measurements and G0/G1 DNA-content density profiles were linked to the corresponding passage when available. The current fit objects contain 131 entries, including 114 with finite growth estimates, 12 with chromosome-number measurements (220 single-cell values in total), and 20 with matched G0/G1 profiles. Each G0/G1 profile was represented on a 200-point ploidy grid and normalized before comparison with the simulated viable chromosome-number distribution.

Within the passage-structured model, oxygen was held fixed at the assigned value for each segment. When a simulated segment was represented by parallel O1 and O2 records, passage duration and initial/final cell counts were summarized by the median, while all available growth, chromosome-number, and flow-density observations contributed to their corresponding likelihood terms.

Supplementary Table 3: Final karyotype distributions of the parallel 2N oxygen-deprivation lines and their shared fitted state. O1 and O2 were propagated as separate lines from the same 2N precursor through the matched nominal oxygen schedule. The fitted row is the single model distribution evaluated against both observed lines at the final matched chromosome-count segment.

Distribution	n	Mean N	Median N	$N \geq 80$	Component structure
O1 observed	20	66.85	49	8/20 (40.0%)	12 cells at $N = 46$ –49; 8 cells at $N = 90$ –100
O2 observed	20	88.05	95	16/20 (80.0%)	4 cells at $N = 49$ –76; 16 cells at $N = 93$ –99
Shared fitted state	—	64.27	60	32.5% probability	dominant local modes at $N = 40$ and $N = 83$

Note: $N \geq 80$ is a descriptive bin chosen within the unoccupied interval between the low- and high-chromosome components of the final observed samples; it is not a fitted threshold or inferential cutoff. Observed entries are single-cell counts. The fitted entry is a probability distribution and therefore has no observed sample size. Values are descriptive and do not constitute an additional statistical test.

1.3 A Mechanistic Model of Chromosome-Number Evolution

We model tumor population dynamics as a deterministic state-structured ODE system over chromosome-number classes. The formulation separates viable cells from non-viable biomass, couples local oxygen availability to resource-dependent growth and stress-associated death, and uses chromosome missegregation and whole-genome doubling to move viable mass across chromosome-number states.

State space and compartments. We consider tumor cells stratified by modeled total autosomal chromosome count; sex chromosomes are not included in this state variable. Let

$$\mathcal{N} = \{N_{\min}, \dots, N_{\max}\}. \quad (1)$$

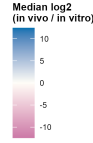
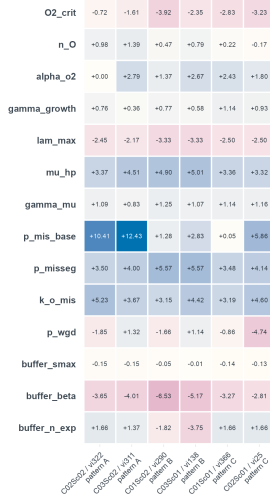
This set denotes the allowed autosomal chromosome-count grid. For the simulations described here, $N_{\min} = 22$ and $N_{\max} = 154$. Let $N_{\text{dip}} = 44$ denote the diploid autosomal chromosome-count reference state, and let $N_{\text{unit}} = 22$ denote the one-ploidy autosomal chromosome-count unit, so that $2N_{\text{unit}} = N_{\text{dip}}$.

Joint-fit scalar parameter contrasts across six warm-start pairs

Pair-level medians and cross-pair directional agreement are optimizer-derived summaries, not biological confidence intervals.

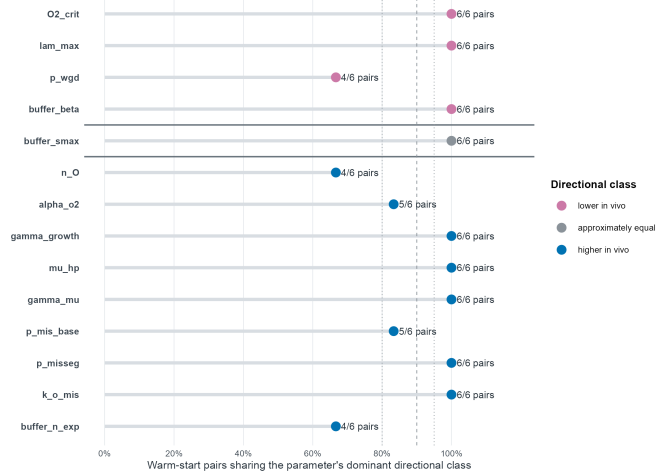
A Pair-level parameter-ratio map

Each cell is the median across 500 numerical seeds within one warm-start pair



B Between-pair directional consensus

Category blocks: lower in vivo ($0 < \text{ratio} \leq 0.8$), approximately equal ($0.8 < \text{ratio} < 1.2$), higher in vivo ($\text{ratio} \geq 1.2$)



A: Pink indicates lower in vivo and blue indicates higher in vivo; trajectory-pattern labels are descriptive model-output categories. B: The warm-start pair is the summarization unit; one pair changes the fraction by 16.7 percentage points.

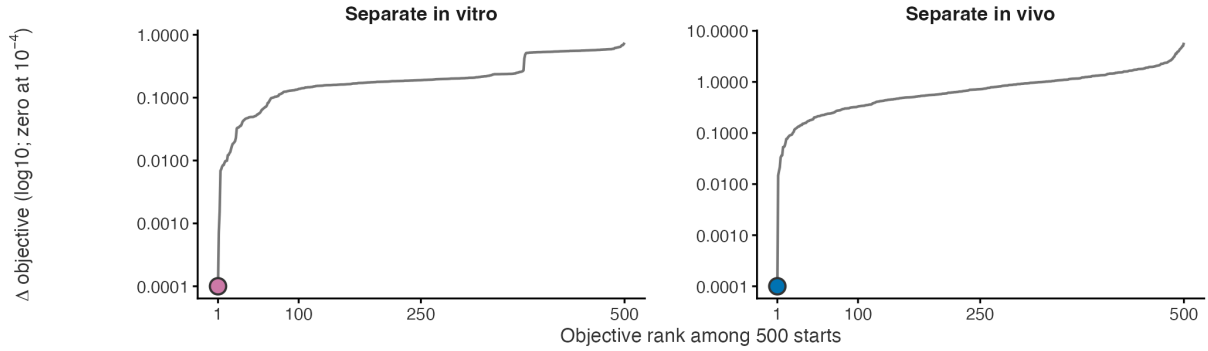
Supplementary Figure 2: **Scalar joint-fit parameter contrasts across the six warm-start pairs.** (A) Pair-by-parameter map of the median $\log_2(\text{in vivo}/\text{in vitro})$ ratio across the 500 numerical seeds within each warm-start pair. Pink indicates lower fitted values *in vivo*, blue indicates higher fitted values *in vivo*, and the printed signed values give the exact pair medians. Pairs are grouped by their final descriptive 0–1,000-day modeled ploidy-trajectory pattern; these are model-output patterns rather than biological groups, and all six pairs share the same seed-10 *in vitro* anchor. (B) Directional consensus across warm-start pairs under the prespecified reclassification used here: lower *in vivo* for $0 < \text{in vivo}/\text{in vitro} \leq 0.8$, approximately equal for $0.8 < \text{in vivo}/\text{in vitro} < 1.2$, and higher *in vivo* for ratios ≥ 1.2 . Parameters are arranged in these three directional blocks rather than sorted by consensus magnitude. Horizontal position and labels show the fraction and exact number of the six pairs sharing the parameter's dominant class. The pair, not the 3,000 numerical runs or biological specimens, is the summarization unit.

Fit-quality landscapes and optimizer diagnostics

A

Separate-fit objective landscapes

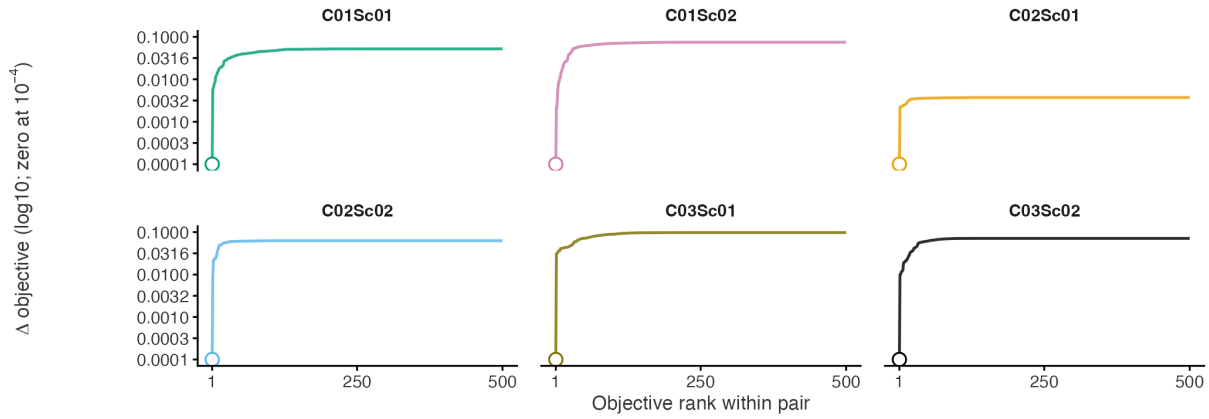
Log₁₀ y-axis; exact zero is displayed at 10^{-4} . Selected fits are highlighted.



B

Joint-fit objective landscapes

Log₁₀ y-axis; exact zero is displayed at 10^{-4} . Facets keep the six rank-1 selected winners distinct.



C

Optimizer diagnostics and solution multiplicity

Flags and optimizer codes are run diagnostics, not confidence intervals; none of these fits has bootstrap, profile-likelihood, or held-out uncertainty.

Separate in vitro	500	0 / 500	accepted; code 1	12 within 0.01	5 / 20
Separate in vivo	500	500 / 500	accepted; code 52	29 within 1% 241 within 5%	2 / 20
Joint: six pair runs	6 x 500	500 / 500 each	0 / 6 accepted	winner totals 18.85–19.98	2–4 / 20
	Starts	DEoptim converged	Local step	Competitive solutions	At bounds

Supplementary Figure 3: **Objective landscapes and optimizer diagnostics for the retained separate and joint fits.** (A) Objective rank landscapes for the 500 separate *in vitro* and 500 separate *in vivo* starts. The y-axis is the objective difference from the lowest retained value on a log₁₀ scale; exact zero is displayed at 10^{-4} . The selected seed-10 *in vitro* and seed-25 *in vivo* fits are highlighted in the context colors. (B) Corresponding objective-difference landscapes for the six 500-start joint warm-start pairs, with the lowest retained winner in each pair highlighted. (C) Recorded run diagnostics for the selected separate fits and six selected joint winners, including attempted starts, differential-evolution convergence flags, local-step status, numbers of competitive solutions, and active-bound counts. Optimizer codes and numerical-start spread diagnose search behavior; they are not biological replication, confidence intervals, bootstrap uncertainty, profile likelihoods, or held-out validation.

Supplementary Table 4: Modeled 4N chromosome-number decline and direct hypoxia-origin dead burden across the ten lowest-objective separate *in vitro* fits. Initial and terminal mean chromosome numbers refer to the first and last modeled states of the selected 4N lineage; its terminal portion was maintained at target 0% O₂. The last column is the maximum daily direct hypoxia-origin dead burden divided by total burden over the full modeled 4N trajectory.

Rank	Fit	Objective	Initial mean N	Terminal mean N	Decline in mean N	Maximum direct hypoxia-origin dead burden (%)
1	seed10	3.8525	84.34	80.76	3.58	1.76
2	seed132	3.8533	84.21	80.66	3.55	1.76
3	seed81	3.8541	84.22	80.67	3.54	1.77
4	seed294	3.8594	84.71	80.89	3.82	1.74
5	seed337	3.8598	84.66	80.76	3.91	1.74
6	seed106	3.8605	84.45	80.53	3.91	1.82
7	seed317	3.8610	84.30	80.13	4.17	1.76
8	seed140	3.8610	84.21	80.54	3.67	1.74
9	seed285	3.8618	84.24	80.20	4.04	1.77
10	seed464	3.8623	84.68	80.73	3.96	1.77

Note: Fits are ordered by the retained objective value. The decline is the initial minus terminal predicted mean chromosome number. The dead-burden percentage is the maximum, over all daily 4N states, of $100 \times \text{burden_dead_hypoxia}/\text{burden_total}$; it is therefore a trajectory maximum rather than an endpoint-only value. These optimizer-selected solutions are not biological replicates or a confidence interval.

The viable abundance vector is

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N \in \mathcal{N}}, \quad (2)$$

where $x_N^L(t)$ is the number of viable cells with modeled autosomal chromosome count N at time t . The ODE state is indexed only by autosomal chromosome count and by live/dead compartment. Unless explicitly marked as a dead compartment, chromosome-number distributions in the model are defined over the viable population. The total viable population size is

$$N_{\text{tot}}(t) = \sum_{N \in \mathcal{N}} x_N^L(t). \quad (3)$$

To represent non-viable biomass contributing to macroscopic tumor volume, we track two dead pools:

$$\mathbf{x}^{D,\text{hyp}}(t) = \{x_N^{D,\text{hyp}}(t)\}_{N \in \mathcal{N}}, \quad \mathbf{x}^{D,\text{CIN}}(t) = \{x_N^{D,\text{CIN}}(t)\}_{N \in \mathcal{N}}. \quad (4)$$

Here $\mathbf{x}^{D,\text{hyp}}$ receives cells lost through stress-associated death, and $\mathbf{x}^{D,\text{CIN}}$ receives non-viable daughter mass generated by chromosome missegregation or WGD; the superscript CIN denotes missegregation-origin loss or out of grid WGD. Their sum is

$$\mathbf{x}^D(t) = \mathbf{x}^{D,\text{hyp}}(t) + \mathbf{x}^{D,\text{CIN}}(t). \quad (5)$$

The model's bulk tumor burden is therefore

$$N_{\text{burden}}(t) = \sum_{N \in \mathcal{N}} \left(x_N^L(t) + x_N^{D,\text{hyp}}(t) + x_N^{D,\text{CIN}}(t) \right), \quad (6)$$

which includes viable cells and dead biomass. This distinction is used throughout: tumor-volume observations are compared with total burden, whereas single-cell chromosome-number observations sample the viable distribution.

Environmental stress functions. Let $O_2(t)$ denote the mean-field approximation of the effective oxygen level experienced by the tumor population. Oxygen/resource stress is represented by the bounded Hill function

$$h(O_2) = \frac{O_{2,c}^{n_O}}{O_{2,c}^{n_O} + O_2^{n_O}}, \quad (7)$$

where $O_{2,c}$ is the critical oxygen level at which the stress term is half-maximal, and n_O controls the steepness of the transition from low stress at high oxygen to strong stress under oxygen limitation. A Hill-type stress function is biologically justified because the literature indicates that oxygen-dependent cell fate is graded but threshold-like rather than linear: “The severity of hypoxia determines whether cells become apoptotic or adapt to hypoxia and survive”,²⁶ and recent spheroid work identifies a “critical survival pO₂”;²⁷ accordingly, Eq. (7) provides a smooth, bounded approximation to an oxygen-linked resource-stress threshold.

Proliferation and stress-associated death rates. Given chromosome count N and local oxygen level O_2 , the effective division rate is

$$\lambda_{\text{eff}}(N, O_2) = \lambda_{\text{max}} \frac{1}{1 + \alpha_{O_2} h(O_2) \left(\frac{N}{N_{\text{dip}}} \right)^{\gamma_{\text{growth}}}}. \quad (8)$$

Here α_{O_2} controls the strength of resource-stress-dependent growth damping, and γ_{growth} controls how strongly this damping increases with chromosome content. When oxygen-linked stress is small, $h(O_2) \approx 0$ and the reciprocal factor approaches one; as oxygen decreases, the reciprocal factor suppresses proliferation more strongly, especially for high-chromosome-count states.

Stress-associated death is applied as an explicit live-to-dead flow. The per-capita death hazard is

$$\mu_{\text{eff}}(N, O_2) = \mu_{\text{hp}} h(O_2) \left(\frac{N}{N_{\text{dip}}} \right)^{\gamma_{\mu}}. \quad (9)$$

Thus μ_{hp} sets the stress-associated death scale, $h(O_2)$ makes death depend on the same oxygen-linked stress function (Eq. (7)), and γ_{μ} controls the chromosome-number penalty on stress-associated death.

Cell division, WGD, and chromosomal transitions. With the stress-modulated rates defined, divisions are represented by a live-state generator matrix

$$G_{\text{div}}(O_2) \in \mathbb{R}^{R \times R}, \quad R = N_{\text{max}} - N_{\text{min}} + 1. \quad (10)$$

Let

$$L(O_2) = \text{diag}(\lambda_{\text{eff}}(N, O_2)) \quad (11)$$

be the diagonal matrix of state-specific division rates. A division follows the ordinary chromosome-missegregation branch with probability $1 - p_{\text{wgd}}$ and follows a whole-genome-doubling branch with

probability p_{wgd} . The corresponding division generator is

$$G_{\text{div}}(O_2) = (1 - p_{\text{wgd}}) B(O_2) L(O_2) + p_{\text{wgd}} W L(O_2) - L(O_2), \quad (12)$$

where $B(O_2)$ is the viable offspring-allocation matrix generated by chromosome missegregation and post-missegregation viability filtering, W maps a mother state N to the within-grid doubled state $2N$, and the final $-L(O_2)$ term removes the dividing mother cell. The ordinary branch B accounts for the two daughter branches of a division and therefore has total mass up to two before viability losses. By contrast, the WGD branch is implemented as an effective tracked-lineage conversion: W has unit mass in the doubled state, so a WGD event replaces the mother lineage by one viable doubled lineage when $2N$ lies in the modeled grid. Out-of-grid WGD mass is not retained in the viable state space and is routed to the CIN-associated dead compartment.

Chromosome missegregation and daughter viability. For a mother cell with chromosome count N , the per-chromosome missegregation probability increases with the stress-associated death hazard defined in Eq. (9):

$$p_{\text{mis}}(N, O_2) = p_{\text{mis}, \text{base}} + p_{\text{misseg}} \frac{\mu_{\text{eff}}(N, O_2)}{\mu_{\text{eff}}(N, O_2) + k_{o, \text{mis}}}, \quad (13)$$

with values clipped to $[0, 1]$. Here $p_{\text{mis}, \text{base}}$ is the baseline per-chromosome missegregation probability under low stress, p_{misseg} is the maximal stress-induced increment above baseline, and $k_{o, \text{mis}}$ is the death-hazard scale at which the stress-induced increase in missegregation is half-maximal.

Let m denote the number of missegregated chromosome copies in a division of a mother cell with chromosome count N . We model

$$m \sim \text{Binomial}\{N, p_{\text{mis}}(N, O_2)\}. \quad (14)$$

For each value of m , the two daughter branches are assigned coarse chromosome-count shifts $+m$ and $-m$. Before applying the daughter-viability modifier, the shift kernel is

$$d_N(\Delta; O_2) = \sum_{m=0}^N \binom{N}{m} \{p_{\text{mis}}(N, O_2)\}^m \{1 - p_{\text{mis}}(N, O_2)\}^{N-m} [\mathbf{1}\{\Delta = m\} + \mathbf{1}\{\Delta = -m\}]. \quad (15)$$

This kernel sums over two daughter branches, so its total mass is two before viability filtering.

For a daughter with chromosome-count shift Δ , post-missegregation survival is approximated by the symmetric ploidy-dependent modifier

$$S_N(\Delta) = s_N^{|\Delta|}, \quad s_N = s_{\text{max}} \exp \left[-\beta_{\text{buf}} \left(\frac{N_{\text{dip}}}{N} \right)^{n_{\text{exp}}} \right]. \quad (16)$$

The parameter s_{max} is the maximum per-copy survival factor, β_{buf} controls the severity of post-missegregation viability loss at lower ploidy, and n_{exp} controls the steepness of the ploidy dependence. This form lets higher-chromosome-count states tolerate a given chromosome-count shift more readily than lower-chromosome-count states. This survival modifier enters the retained-grid offspring allocation and dead-pool accounting described in the boundary-handling subsection below.

Boundary handling on the chromosome-count domain. The chromosome-count state space is finite. We define the viable chromosome-count domain as

$$\mathcal{N} = \{N_{\min}, N_{\min} + 1, \dots, N_{\max}\},$$

where $N_{\min} = 22$ and $N_{\max} = 154$ in the simulations described here. The viable state vector

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N \in \mathcal{N}}$$

therefore tracks only cells whose modeled autosomal chromosome count lies within this retained grid.

This finite state space requires an explicit rule for division events that would produce daughter or doubled-lineage states outside $\mathcal{N}$. We use absorbing boundary handling on the viable chromosome-count grid. That is, daughter cells or WGD-derived lineages whose chromosome count would fall below $N_{\min}$ or exceed $N_{\max}$ are removed from the viable offspring allocation and routed to the CIN-associated dead compartment.

For ordinary missegregating divisions, the viable offspring-allocation matrix is defined only over retained daughter states $N' \in \mathcal{N}$:

$$B_{N',N}(O_2) = d_N(N' - N; O_2) S_N(N' - N), \quad N, N' \in \mathcal{N}.$$

Here $d_N(\Delta; O_2)$ specifies the probability of generating a daughter chromosome-count shift Δ , and $S_N(\Delta)$ specifies the probability that a daughter with that shift remains viable. Any probability mass corresponding to $N + \Delta \notin \mathcal{N}$ is excluded from $B_{\cdot,N}$. Similarly, for whole-genome-doubling events, the doubling operator W retains only within-grid transitions $N \mapsto 2N \in \mathcal{N}$; out-of-grid WGD offspring mass is excluded from $W_{\cdot,N}$ and routed to the CIN-associated dead compartment.

Because the viable offspring allocation can be subunitary, either because buffering reduces daughter survival or because daughter or doubled-lineage states fall outside the retained chromosome-count grid, we define the CIN-associated event-level nonviable inflow as

$$\phi_{\text{CIN},N}(\mathbf{x}^L, O_2) = \lambda_{\text{eff}}(N, O_2) x_N^L \left\{ (1 - p_{\text{wgd}}) \left[2 - \sum_{N' \in \mathcal{N}} B_{N',N}(O_2) \right] + p_{\text{wgd}} \left[1 - \sum_{N' \in \mathcal{N}} W_{N',N} \right] \right\}. \quad (17)$$

The first bracket is the ordinary-division deficit between two potential daughters and the viable daughter mass retained within $\mathcal{N}$. The second bracket accounts for WGD events for which the doubled state $2N$ lies outside the modeled grid. This term is distinct from the continuous hypoxia-associated death hazard $\mu_{\text{eff}}(N, O_2)$; it is added to the CIN-associated dead compartment $x^{D,\text{CIN}}$.

Thus, the chromosome-count boundaries are absorbing for viable offspring mass, with out-of-domain and non-buffered daughter mass transferred to the dead compartment.

Oxygen demand and supply dynamics. To couple tumor size to oxygen demand, we define the chromosome-number-weighted oxygen-demand proxy

$$N_{\text{eff}}(t) = \sum_N x_N^L(t) \left(\frac{N}{N_{\text{dip}}} \right)^\eta, \quad (18)$$

where η controls how oxygen demand scales with chromosome content. The reference scale N_{ref} is a fixed cell-count scale used to nondimensionalize demand, so that $N_{\text{eff}}/N_{\text{ref}}$ is a relative effective tumor size.

The oxygen supply target is the phenomenological supply–demand relation

$$O_2^*(N_{\text{eff}}) = \max\left(O_{2,\min}, S_0 - \kappa_O \log\left(1 + \frac{N_{\text{eff}}}{N_{\text{ref}}}\right)\right), \quad (19)$$

where S_0 is the low-burden oxygen supply level, $O_{2,\min}$ is the oxygen floor, and κ_O controls the oxygen-drop amplitude per unit log-burden increase. The logarithmic term is dimensionless because of the normalization by N_{ref} . Oxygen values are interpreted as percentages and are restricted to the physical interval $[0, 100]$.

Effective oxygen is updated by first-order relaxation toward this target:

$$\alpha_\tau = 1 - \exp\left(-\frac{\Delta t}{\tau_{O_2}}\right), \quad O_{2,k+1} = O_{2,k} + \alpha_\tau (O_2^*(N_{\text{eff},k}) - O_{2,k}). \quad (20)$$

Thus oxygen changes gradually toward the demand-adjusted target rather than instantaneously matching it at each simulation step.

Discrete-time simulation scheme. The simulations use a time step of $\Delta t = 0.05$ days. At each step $k \rightarrow k+1$, the oxygen state is first relaxed toward the supply–demand target in Eq. (20), using the current viable population to calculate $N_{\text{eff},k}$. The resulting relaxed value, $O_{2,k+1}$, is then used to evaluate division-linked chromosome transitions and explicit stress-associated death for that population step:

$$\mathbf{x}_{k+1}^L = \mathbf{x}_k^L + \Delta t G_{\text{div}}(O_{2,k+1})\mathbf{x}_k^L - \Delta t \mu_{\text{eff}}(N, O_{2,k+1}) \odot \mathbf{x}_k^L, \quad (21)$$

where $\odot$ denotes element-wise multiplication over chromosome-count states.

Mass that becomes nonviable due to hypoxia or failed divisions is transferred into the dead compartments, which contribute to tumor burden. The resource-stress-origin dead pool is updated as

$$\mathbf{x}_{k+1}^{D,\text{hyp}} = \mathbf{x}_k^{D,\text{hyp}} + \Delta t \mu_{\text{eff}}(N, O_{2,k+1}) \odot \mathbf{x}_k^L - \Delta t k_{\text{clear}} \mathbf{x}_k^{D,\text{hyp}}, \quad (22)$$

and the CIN-associated dead pool is updated as

$$\mathbf{x}_{k+1}^{D,\text{CIN}} = \mathbf{x}_k^{D,\text{CIN}} + \Delta t \phi_{\text{CIN}}(\mathbf{x}_k^L, O_{2,k+1}) - \Delta t k_{\text{clear}} \mathbf{x}_k^{D,\text{CIN}}. \quad (23)$$

Here k_{clear} is the first-order clearance rate for dead biomass, and ϕ_{CIN} is defined component-wise in Eq. (17). Together, the oxygen relaxation and Eqs. (21)–(23) define the coupled viable/dead population dynamics used in the simulations.

1.3.1 Model calibration and observation models

Model calibration uses a common transformed-parameter optimizer for *in vivo*, *in vitro*, and joint fits. Natural-scale parameter specifications provide initialization values and admissible bounds; each component maps its fitted parameters to the optimizer scale before evaluating its objective.

The mechanistic dynamics described above are shared, while the observation models differ between the *in vivo* and *in vitro* data.

***In vivo* observation model and likelihood.** The *in vivo* component uses three data modalities: longitudinal tumor burden, terminal necrosis where matched histology is available, and terminal single-cell chromosome-number profiles.

Tumor burden. For each mouse $i \in \mathcal{I}$, where $\mathcal{I}$ denotes the eight retained tumors used for calibration (section 1.1), tumor burden is measured at observation times $\{t_{ij}^{(V)}\}_{j=1}^{m_i}$, yielding observed tumor volumes

$$V_{ij}^{\text{obs}} \equiv V_i^{\text{obs}}(t_{ij}^{(V)}). \quad (24)$$

Let $x_{i,N}^L(t)$, $x_{i,N}^{D,\text{hyp}}(t)$, and $x_{i,N}^{D,\text{CIN}}(t)$ denote the viable, hypoxia-origin dead, and CIN-associated dead abundances for tumor i in chromosome-count state N at time t . The model predicts tumor volume by scaling total burden:

$$\widehat{V}_{ij} = \sum_N \left(x_{i,N}^L(t_{ij}^{(V)}) + x_{i,N}^{D,\text{hyp}}(t_{ij}^{(V)}) + x_{i,N}^{D,\text{CIN}}(t_{ij}^{(V)}) \right) v_{\text{cell}}, \quad (25)$$

where each cell is assigned the effective volume

$$v_{\text{cell}} = \frac{1}{\rho_{2N}}. \quad (26)$$

This observation model does not impose a ploidy-dependent cell-volume correction: low- and high-ploidy cells contribute the same effective per-cell volume to the bulk burden. Tumor-burden observations are modeled on the logarithmic scale,

$$\log V_{ij}^{\text{obs}} \sim \mathcal{N}\left(\log \widehat{V}_{ij}, \sigma_{\text{burden}}^2\right), \quad (27)$$

where σ_{burden} is the burden measurement-noise scale. For longitudinal tumor burden, the fitted loss is a Gaussian negative log-likelihood on the logarithmic scale, averaged first within tumor and then across tumors. Let $\mathcal{J}_i$ denote the retained observation times for tumor i after excluding day 0. Then

$$\mathcal{L}_B(\theta) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|\mathcal{J}_i|} \sum_{j \in \mathcal{J}_i} \left[\log \sigma_{\text{burden}} + \frac{1}{2} \log(2\pi) + \frac{\left(\log V_{ij}^{\text{obs}} - \log \widehat{V}_{ij}\right)^2}{2\sigma_{\text{burden}}^2} \right]. \quad (28)$$

In implementation, each volume entering a logarithm is replaced by $\max(V, \varepsilon_{\log B})$ with $\varepsilon_{\log B} = 10^{-12} \text{ mm}^3$; this numerical floor is suppressed in Eqs. (27)–(28) for readability. Because σ_{burden} is estimated jointly, the $\log \sigma_{\text{burden}}$ term is retained in the likelihood contribution. Omitting it would allow the optimizer to reduce the burden loss by inflating the assumed measurement noise.

Terminal necrosis. For tumors with matched terminal histology, necrosis is compared with the simulated terminal fraction of total dead biomass. This is an observation-model term, not an additional live-to-dead flux. The stress-associated death hazard $\mu_{\text{eff}}(N, O_2)$ contributes to $\mathbf{x}^{D,\text{hyp}}$, while mitotic nonviability and boundary-dropped offspring contribute to $\mathbf{x}^{D,\text{CIN}}$. The terminal necrosis comparison uses the sum of these two dead pools.

For tumor i , define the terminal total-dead volume and terminal total tumor volume as

$$\widehat{V}_i^{D,\text{tot}}(\theta) = v_{\text{cell}} \sum_N \left[x_{i,N}^{D,\text{hyp}}(T_i^{(H)}; \theta) + x_{i,N}^{D,\text{CIN}}(T_i^{(H)}; \theta) \right], \quad \widehat{V}_i^{\text{tot}}(\theta) = v_{\text{cell}} \sum_N \left[x_{i,N}^L(T_i^{(H)}; \theta) + x_{i,N}^{D,\text{hyp}}(T_i^{(H)}; \theta) + \right. \\ \left. x_{i,N}^{D,\text{CIN}}(T_i^{(H)}; \theta) \right] \quad (29)$$

The predicted terminal necrotic fraction is

$$\widehat{f}_i^{\text{nec}}(\theta) = \frac{\widehat{V}_i^{D,\text{tot}}(\theta)}{\widehat{V}_i^{\text{tot}}(\theta)}. \quad (30)$$

Because the same effective cell-volume factor appears in both numerator and denominator, it cancels in this ratio. Thus the fitted necrosis term constrains the terminal total-dead fraction of the simulated tumor burden, after both dead pools have accumulated their respective inflows and have been cleared by the shared first-order clearance process.

Let $f_i^{\text{nec,obs}}$ be the observed terminal necrotic fraction for tumors in the matched necrosis set $\mathcal{I}_{\text{nec}}$. Fractions are compared on a clipped logit scale. Define

$$\text{logit}_\epsilon(f) = \log \left(\frac{\tilde{f}}{1 - \tilde{f}} \right), \quad \tilde{f} = \min\{1 - \epsilon_{\text{nec}}, \max(\epsilon_{\text{nec}}, f)\}. \quad (31)$$

The terminal necrosis loss is the mean standardized squared discrepancy on this scale,

$$\mathcal{L}_{\text{nec}}(\theta) = \frac{1}{|\mathcal{I}_{\text{nec}}|} \sum_{i \in \mathcal{I}_{\text{nec}}} \left[\frac{\text{logit}_\epsilon(\widehat{f}_i^{\text{nec}}(\theta)) - \text{logit}_\epsilon(f_i^{\text{nec,obs}})}{\sigma_{\text{nec}}} \right]^2. \quad (32)$$

Here σ_{nec} sets the tolerated discrepancy in terminal necrosis on the logit-fraction scale, and ϵ_{nec} prevents fractions at 0 or 1 from producing infinite logits. In the run configuration these quantities are named `sigma_necrosis_logit` and `necrosis_fraction_eps`, respectively.

Ploidy. Each mouse has an initial ploidy group $g_i \in \{2N, 4N\}$ and harvest time $T_i^{(H)}$. Retained burden observations are nonzero-time measurements up to harvest, so the number and timing of observations may differ across mice.

For each tumor $i \in \mathcal{I}$, endpoint single-cell copy-number profiling provides modeled total autosomal chromosome-number estimates. If n_i cells are profiled from tumor i , the observed chromosome-number values are denoted

$$z_{i\ell}, \quad \ell = 1, \dots, n_i. \quad (33)$$

These single-cell observations sample the viable population at harvest. The model-predicted viable chromosome-number distribution at harvest is

$$\widehat{\pi}_{i,N}(\theta) = \frac{x_{i,N}^L(T_i^{(H)}; \theta)}{\sum_{N'} x_{i,N'}^L(T_i^{(H)}; \theta)}. \quad (34)$$

This distribution represents the predicted probability that a viable cell sampled from tumor i at harvest has true chromosome count N .

Observed single-cell chromosome-number estimates are then modeled conditional on this predicted viable distribution. Specifically, each observed value $z_{i\ell}$ is treated as a noisy measurement

of an unobserved true chromosome-count state N , with mixture weights given by $\hat{\pi}_{i,N}(\theta)$:

$$p(z_{i\ell} \mid \theta) = \sum_N \hat{\pi}_{i,N}(\theta) \mathcal{N}(z_{i\ell}; N, \sigma_{\text{ploidy}}^2). \quad (35)$$

where σ_{ploidy} represents measurement variability in single-cell chromosome-number estimates. For terminal single-cell chromosome-number data, the fitted loss first computes a tumor-level average negative log-likelihood. Define

$$\ell_i(\theta) = -\frac{1}{n_i} \sum_{\ell=1}^{n_i} \log p(z_{i\ell} \mid \theta). \quad (36)$$

Let $\mathcal{I}_{2N}$ and $\mathcal{I}_{4N}$ denote the retained tumors initiated from the near-diploid and near-tetraploid lineages, respectively. With both cohorts present, the endpoint contribution gives equal weight to the two initial-ploidy cohorts and equal weight to tumors within each cohort:

$$\mathcal{L}_C(\theta) = \frac{1}{2} \left[\frac{1}{|\mathcal{I}_{2N}|} \sum_{i \in \mathcal{I}_{2N}} \ell_i(\theta) + \frac{1}{|\mathcal{I}_{4N}|} \sum_{i \in \mathcal{I}_{4N}} \ell_i(\theta) \right]. \quad (37)$$

Maximum-a-posteriori objective. The *in vivo* component is formulated as a maximum-a-posteriori objective combining burden likelihood, chromosome-number likelihood, terminal necrosis loss, and soft-prior penalties:

$$\mathcal{L}_{\text{vivo}}(\theta) = \mathcal{L}_B(\theta) + \mathcal{L}_C(\theta) + \lambda_{\text{nec}} \mathcal{L}_{\text{nec}}(\theta) + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}}(\theta), \quad (38)$$

where θ denotes the fitted parameter vector for this component, $\mathcal{L}_B$ is the burden contribution, $\mathcal{L}_C$ is the terminal chromosome-number contribution, $\mathcal{L}_{\text{nec}}$ is the terminal total-dead-fraction loss, λ_{nec} is its relative objective weight, and $\mathcal{L}_{\text{prior}}$ is the soft-prior penalty.

For each prior-constrained parameter, let $z_i = T_i(\theta_i)$ denote the calibration-scale value used by the optimizer. The soft-prior penalty is

$$\mathcal{L}_{\text{prior}}(\theta) = \frac{1}{2} \sum_{i \in \mathcal{P}} \left(\frac{z_i - \mu_i}{\sigma_i} \right)^2, \quad (39)$$

where $\mathcal{P}$ is the subset of parameters with configured soft priors. The transform T_i is the parameter-specific optimizer transform (including $\log_{10}$ transforms for several positive parameters and the identity transform where configured), and μ_i and σ_i are the prior center and scale on that same calibration scale.

***In vitro* observation model and likelihood.** The *in vitro* component uses the same chromosome-number transition model but applies it to passage-structured cell-line data. Each passage segment s has an initial cohort label, a fixed oxygen level during the segment, and a duration Δt_s ; available growth observations are represented as segment-level growth-rate data rather than being restricted to a single initial-final cell-count pair. The model propagates the corresponding viable chromosome-number distribution through the segment and compares the simulated passage outcome with three measured data modalities: passage growth-rate observations, passage-level chromosome-number ob-

servations, and flow-density measurements. For each passage, the propagated chromosome-number distribution used for reseeded and endpoint comparison was taken from the simulated time point whose live-cell count was closest to the observed final or next-passage initial cell count.

For growth-rate data, let g_s^{obs} be the observed passage growth rate and $\hat{g}_s(\theta)$ the model-predicted growth rate for segment s . The averaged growth log-likelihood is

$$\bar{\ell}_{\text{growth}}(\theta) = \frac{1}{|\mathcal{S}_g|} \sum_{s \in \mathcal{S}_g} \log \mathcal{N}(g_s^{\text{obs}}; \hat{g}_s(\theta), \sigma_{\text{growth}}^2), \quad (40)$$

where $\mathcal{S}_g$ is the set of segments with observed growth-rate data and σ_{growth} is the growth measurement-noise scale.

For chromosome-number observations in segment s , let $\pi_{s,N}$ denote the predicted viable chromosome-number distribution at the comparison time and let $z_{s\ell}$ be the observed integer chromosome-number value for cell ℓ . The implementation first smooths the predicted distribution on the finite modeled chromosome grid $\mathcal{N}$. For an observed-grid state m and true modeled state N , define the column-normalized Gaussian kernel

$$K_{\sigma_{\text{kary}}}(m | N) = \frac{\phi(m; N, \sigma_{\text{kary}}^2)}{\sum_{r \in \mathcal{N}} \phi(r; N, \sigma_{\text{kary}}^2)}, \quad m, N \in \mathcal{N}, \quad (41)$$

where $\phi(\cdot; N, \sigma_{\text{kary}}^2)$ is a Gaussian density evaluated only on the discrete chromosome grid. The smoothed discrete probability mass is

$$\tilde{\pi}_{s,m} = \sum_{N \in \mathcal{N}} K_{\sigma_{\text{kary}}}(m | N) \pi_{s,N}. \quad (42)$$

The averaged chromosome-number log-likelihood is then

$$\bar{\ell}_{\text{ploidy}}(\theta) = \frac{1}{|\mathcal{S}_z|} \sum_{s \in \mathcal{S}_z} \frac{1}{n_s} \sum_{\ell=1}^{n_s} \log [\max \{ \tilde{\pi}_{s,z_{s\ell}}, 10^{-12} \}], \quad (43)$$

where $\mathcal{S}_z$ is the set of passage comparisons with chromosome-number observations, n_s is the number of observations in segment s , and σ_{kary} is the chromosome-number smoothing scale for the *in vitro* karyotype term. Observations that do not match a modeled grid state receive the same 10^{-12} probability floor. Thus, this term is a discrete grid likelihood after finite-grid kernel smoothing, not a continuous Gaussian-mixture density.

For flow-density data, let ω_{sp}^{obs} be the normalized observed density mass at flow grid point p for segment s , and let $\hat{f}_s(p; \theta)$ be the model-predicted density after Gaussian smoothing of the simulated viable chromosome-state distribution onto the observed flow grid and renormalization over that grid. The Gaussian width was set separately from the fitted karyotype smoothing parameter to the larger of twice the median flow-grid spacing and 1/22 ploidy units; for the seed-10 grid this was 0.0804020100502587. The averaged flow log-likelihood is

$$\bar{\ell}_{\text{flow}}(\theta) = \frac{1}{|\mathcal{S}_f|} \sum_{s \in \mathcal{S}_f} \sum_{p \in \mathcal{G}_s} \omega_{sp}^{\text{obs}} \log [\max \{ \hat{f}_s(p; \theta), 10^{-12} \}], \quad (44)$$

where $\mathcal{S}_f$ is the set of flow-profile comparisons and $\mathcal{G}_s$ is the observed flow grid for segment s . The 10^{-12} floor is a numerical safeguard applied after smoothing and normalization.

The fitted *in vitro* objective is the negative weighted mean log-likelihood

$$\mathcal{L}_{\text{vitro}}(\theta) = - \left[w_{\text{growth}} \bar{\ell}_{\text{growth}}(\theta) + w_{\text{ploidy}} \bar{\ell}_{\text{ploidy}}(\theta) + w_{\text{flow}} \bar{\ell}_{\text{flow}}(\theta) \right]. \quad (45)$$

The three terms compare the simulated passage lineages with growth-rate, chromosome-number, and flow-density data, respectively. The reported fits used equal component weights, $w_{\text{growth}} = w_{\text{ploidy}} = w_{\text{flow}} = 1$.

Joint fitting of *in vivo* and *in vitro* data. The joint fit uses a single optimizer vector while evaluating separate transformed parameter vectors for the *in vivo* and *in vitro* likelihood components. Soft-coupled parameters are represented by a center-delta parameterization: for each soft-coupled parameter, the optimizer carries one transformed center variable, c_Q , and one additional offset variable, δ_Q . The context-specific transformed values are reconstructed from these two optimizer variables during objective evaluation. Parameters that appear only in one observation model are retained as context-specific observation-model parameters. In the analysis reported here, all 14 overlapping active biological parameters— $O_{2,c}$, μ_{hp} , p_{misseg} , $k_{o,\text{mis}}$, s_{max} , β_{buf} , n_{exp} , n_O , α_{O_2} , γ_{growth} , λ_{max} , $p_{\text{mis,base}}$, p_{wgd} , and γ_{μ} —were soft-coupled across contexts; no overlapping active biological parameter was hard-shared. Thus, the strength and chromosome-number dependence of resource-stress growth suppression were allowed to differ between the *in vivo* and *in vitro* components, while regularization discouraged unsupported divergence.

The minimized joint objective is

$$\mathcal{L}_{\text{joint}}(\theta) = w_{\text{vivo}} \mathcal{L}_{\text{vivo}}(\theta_{\text{vivo}}) + w_{\text{vitro}} \mathcal{L}_{\text{vitro}}(\theta_{\text{vitro}}) + \mathcal{L}_{\text{soft}}(\theta), \quad (46)$$

where θ_{vivo} and θ_{vitro} are the context-specific parameter vectors reconstructed from the joint optimizer vector and passed to the two likelihood components. The reported analysis used $w_{\text{vivo}} = w_{\text{vitro}} = 1$. For each soft-coupled natural-scale parameter Q , let

$$z_Q = T_Q(Q)$$

denote its transformed optimizer-scale value, where T_Q is the transformation used for that parameter during optimization. The separate *in vivo* and *in vitro* fits define context-specific transformed bounds $[l_Q^{\text{vivo}}, u_Q^{\text{vivo}}]$ and $[l_Q^{\text{vitro}}, u_Q^{\text{vitro}}]$. In the joint fit, these component-specific bounds are used only to define a single active joint admissible interval:

$$l_Q^{\text{joint}} = \min(l_Q^{\text{vivo}}, l_Q^{\text{vitro}}), \quad u_Q^{\text{joint}} = \max(u_Q^{\text{vivo}}, u_Q^{\text{vitro}}). \quad (47)$$

Thus, the original component-specific bounds record the provenance of the separate fits, whereas the joint optimizer uses the union interval $[l_Q^{\text{joint}}, u_Q^{\text{joint}}]$ as the active transformed-scale admissible range. During objective evaluation, the transformed values passed to the two components are reconstructed as

$$z_Q^{\text{vivo}} = c_Q + \frac{\delta_Q}{2}, \quad z_Q^{\text{vitro}} = c_Q - \frac{\delta_Q}{2}. \quad (48)$$

The reconstructed values must satisfy

$$z_Q^{\text{vivo}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}], \quad z_Q^{\text{vitro}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}]. \quad (49)$$

Parameter vectors that do not yield admissible reconstructed transformed values are outside the joint feasible domain and are not evaluated as valid likelihood points. The soft-coupling penalty is applied to the offset variable,

$$\mathcal{L}_{\text{soft}}(\theta) = \sum_{Q \in \mathcal{S}} \frac{c^2}{2} \left[1 - \exp \left\{ - \left(\frac{\delta_Q}{c\sigma_Q} \right)^2 \right\} \right], \quad (50)$$

where $\mathcal{S}$ is the set of soft-coupled parameters, σ_Q is the transformed-scale soft-coupling scale, and c controls the transition from near-quadratic behavior to saturation. For the reported fits, $\sigma_Q = 0.65$ for all 14 parameters and $c = 0.4$. For $|\delta_Q| \ll c\sigma_Q$, each term approaches $\frac{1}{2}(\delta_Q/\sigma_Q)^2$; for large separations it saturates at $c^2/2$, limiting the influence of a single strongly context-specific parameter. Since

$$z_Q^{\text{vivo}} - z_Q^{\text{vitro}} = \delta_Q,$$

this penalty regularizes the transformed difference between the *in vivo* and *in vitro* parameter values. For log-transformed parameters, this corresponds to regularization of fold differences rather than absolute natural-scale differences.

Warm-start initialization for the joint fit. The joint analysis used six landscape-informed separate *in vivo* fits as warm-start representatives. The three primary *in vivo* clusters were the $k = 3$ t-SNE-coordinate regions from the pooled 14-parameter joint-initialization landscape described below (sizes 99, 385, and 16; average silhouette 0.816), not the *in vivo*-only 18-parameter clusters in Figure 4B. Within each primary cluster, the same 14 parameter features were first put on their configured natural or $\log_{10}$ scales and then z-scored within that primary cluster. Candidate k-means partitions with $k = 2, \dots, 6$ were evaluated by average silhouette; a primary cluster with fewer than six solutions would not have been subdivided. The k-means seed was $1123 + i - 1$ for primary-cluster iteration i . Maximum silhouette selected $k = 2$ in all three primary clusters, with average silhouettes of 0.179, 0.096, and 0.330. The objective-minimum separate-fit seed from each of the resulting six subclusters was selected as an *in vivo* warm start. These subclusters were used to distribute starting points across the fitted landscape; their low-to-moderate silhouettes do not establish biological subtypes.

Each of the six *in vivo* representatives was paired with the same lowest-objective separate *in vitro* fit (seed 10) on the transformed optimizer scale. Seed 10 belonged to a 376-seed *in vitro* subcluster, and the remaining *in vitro* subclusters were not represented in the joint warm-start panel. Thus, the design explored six *in vivo* regions against one fixed *in vitro* anchor rather than a balanced cross-product of both landscapes.

For each soft-coupled parameter Q , let $z_Q^{\text{vivo},0}$ and $z_Q^{\text{vitro},0}$ denote the transformed values from one such pair. The initial center and offset are set to

$$c_Q^0 = \frac{z_Q^{\text{vivo},0} + z_Q^{\text{vitro},0}}{2}, \quad \delta_Q^0 = z_Q^{\text{vivo},0} - z_Q^{\text{vitro},0}. \quad (51)$$

This initialization exactly reconstructs the two separate-fit transformed values:

$$z_Q^{\text{vivo},0} = c_Q^0 + \frac{\delta_Q^0}{2}, \quad z_Q^{\text{vitro},0} = c_Q^0 - \frac{\delta_Q^0}{2}.$$

Parameters that appear only in one component, such as context-specific observation-model parameters, are initialized from the corresponding separate-fit solution.

The warm start must be representable under the active joint union bounds in Eq. (47); otherwise, initialization fails explicitly. This ensures that the joint fit begins from the selected separate-fit solutions only when those solutions are valid under the declared joint feasible domain.

Optimization and data inclusion. Optimization attempted a parallel differential-evolution global stage followed by serial L-BFGS-B refinement; a local result was retained only when it passed the implementation checks and improved the objective. Separate *in vivo* and *in vitro* searches each used 500 numerical seeds. In the joint analysis, each of the six warm-start pairs used 500 numerical seeds, for 3,000 optimizer runs in total. All 3,000 joint runs recorded an attempted local stage with convergence code 52 and `optimizer_local_accepted=FALSE`; the reported joint solutions are therefore the retained differential-evolution endpoints. Every joint differential-evolution run ended by the configured relative-tolerance/step-tolerance early-stopping rule after 26–116 iterations of the 500-iteration target. The separate *in vitro* convergence summary likewise labeled 0/500 differential-evolution runs as converged; the best retained seed-10 local step was accepted but returned convergence code 1. These diagnostics are summarized in Supplementary Figure 3 and mean that “best” denotes the lowest retained objective among the supplied searches, not a proven global optimum. Seed-to-seed spread was used to assess numerical search robustness and was not interpreted as biological replication or inferential uncertainty.

Retained *in vivo* observation times satisfy $0 < t_{i,1} < \dots < t_{i,n_i} \leq T_i^{(H)}$, so day-0 burden points are excluded and longitudinal follow-up is administratively truncated at harvest. No censoring likelihood was used. Only tumors with both burden and terminal chromosome-number data are retained. Terminal endpoint discrepancies are evaluated directly from the single-cell chromosome-number observations under the Gaussian-mixture likelihood in Eq. (35), and tumor-level endpoint losses are aggregated equally within each initial-ploidy cohort before applying cohort balancing as in Eq. (37).

Representative *in vivo* trajectory and terminal-distribution visualization. The lowest-objective retained separate *in vivo* fit (seed 25) was propagated at one-day reporting intervals for the eight paired tumors. The four 2N-initiated and four 4N-initiated trajectories were summarized separately. At each day, viable, hypoxia-origin dead, and CIN-associated dead model states were converted to tumor-volume components and averaged over the tumor trajectories available on that day before being plotted as stacked areas. The fitted effective oxygen state was plotted directly in percent. Viable-cell mean ploidy was calculated within each tumor as

$$\bar{P}_i(t) = \frac{\sum_{N \in \mathcal{N}} N x_{i,N}^L(t)}{N_{\text{unit}} \sum_{N \in \mathcal{N}} x_{i,N}^L(t)}, \quad N_{\text{unit}} = 22. \quad (52)$$

The tumor-specific effective- O_2 and $\bar{P}_i(t)$ trajectories were then averaged over the same available trajectories at each day. Both cohort means were displayed without standardization on the numerical 0–5 left axis, while burden used the right axis. Sharing a coordinate axis is a visual alignment only and does not imply common physical units. Both longitudinal panels use a 0–100-day x-axis; model curves stop at the last available modeled time rather than being extrapolated to day 100.

Observed tumor burden was summarized only at recorded measurement days. For cohort c , day t , and n_{ct} observed tumors, the black curve is the arithmetic mean $\bar{B}_c^{\text{obs}}(t)$, and the gray band is $\bar{B}_c^{\text{obs}}(t) \pm s_c(t)$, where $s_c(t)$ is the sample standard deviation, truncated at zero for the lower display boundary. The 2N cohort had $n_{ct} = 4$ at every displayed measurement day. The 4N cohort had $n_{ct} = 4$ through day 77 and $n_{ct} = 2$ at days 81–91. The band therefore describes between-tumor dispersion, not a confidence interval.

For the terminal comparison, the observed single-cell chromosome-number values and predicted terminal chromosome-state weights underlying the seed-25 report’s Figure 2.2 were converted to ploidy by division by $N_{\text{unit}} = 22$. Within each cohort and source, each tumor was normalized to unit total weight and then assigned one-quarter of the cohort distribution. This equal-tumor weighting prevents tumors with more profiled cells from dominating the observed cohort distribution. Observed and predicted distributions were displayed on the common ploidy range 1–7 with weighted box summaries.

Fixed-oxygen attractor analysis. For each of the 500 separate *in vivo* parameter sets, the chromosome-number model was evaluated at fixed oxygen without parameter refitting. At each oxygen level, the chromosome-transition generator was combined with the state-specific effective death rate to form

$$M(O_2) = G_{\text{div}}(O_2) - \text{diag}\{\mu_{\text{eff}}(N, O_2)\}. \quad (53)$$

The eigenvalue of M with the largest real part, λ_1 , was calculated numerically, and the real part of its associated eigenvector was retained. The eigenvector sign was oriented to have a non-negative sum, negative numerical components were truncated to zero, and the vector was normalized to unit mass. With normalized dominant eigenvector v_N , dominant mean ploidy was

$$\bar{P}_{\text{dom}}(O_2) = \frac{1}{N_{\text{unit}}} \sum_{N \in \mathcal{N}} N v_N. \quad (54)$$

The spectral gap was defined as the difference between the largest and second-largest eigenvalue real parts, $\text{Re}(\lambda_1) - \text{Re}(\lambda_2)$. For the ploidy-outcome separation analysis, parameter sets with $\bar{P}_{\text{dom}} \geq 2$ were assigned directly to the “dominant mean ploidy ≥ 2 ” class and those with $\bar{P}_{\text{dom}} < 2$ to the “dominant mean ploidy < 2 ” class. The operator was evaluated at 201 fixed oxygen concentrations from 0% through 5% O_2 in increments of 0.025 percentage points. These are analytical evaluations of the fitted linear operator, not stochastic simulations or oxygen-intervention experiments.

Post-fit single-parameter ploidy-outcome separation analysis. At each of the 201 oxygen concentrations, the analysis units were the 500 parameter sets obtained from distinct numerical optimizer seeds, not biological replicates. Each of 18 prespecified fitted *in vivo* parameters was analyzed separately; the observation/volume parameters ρ_{2N} and σ_{burden} were excluded. The parameters s_{max} , γ_{growth} , γ_{μ} , and n_O were retained on their natural scales; the remaining strictly

positive parameters were $\log_{10}$ transformed. Because these transformations are monotone, they do not change the rank-based AUC.

With dominant mean ploidy ≥ 2 coded as $y_i = 1$ and dominant mean ploidy < 2 coded as $y_i = 0$, let R_i denote the average rank of the transformed value of one parameter, and let n_+ and n_- denote the numbers of fits in the two classes. The raw single-parameter receiver-operating-characteristic AUC was

$$\text{AUC}_{\text{raw}} = \frac{\sum_{i:y_i=1} R_i - \frac{n_+(n_++1)}{2}}{n_+n_-}. \quad (55)$$

Equivalently, this quantity is $\Pr(X_+ > X_-) + \frac{1}{2}\Pr(X_+ = X_-)$, where X_+ and X_- are the parameter values from randomly selected fitted solutions in the dominant-mean-ploidy ≥ 2 and < 2 classes, respectively. Thus, a raw AUC above 0.5 indicates that higher parameter values tend to occur in the ≥ 2 class, whereas a raw AUC below 0.5 indicates that higher values tend to occur in the < 2 class.

Direction-independent separation was summarized as

$$\text{AUC}_{\text{disc}} = \max\{\text{AUC}_{\text{raw}}, 1 - \text{AUC}_{\text{raw}}\}. \quad (56)$$

For the heatmap, AUC_{disc} was assigned a positive sign when $\text{AUC}_{\text{raw}} > 0.5$ and a negative sign when $\text{AUC}_{\text{raw}} < 0.5$; red therefore denotes association of higher parameter values with dominant mean ploidy ≥ 2 , and blue denotes association with dominant mean ploidy < 2 . The legend reports only the non-negative discriminative magnitude. For each parameter, the peak-AUC summary retained its maximum AUC_{disc} across the 201-point oxygen grid and the oxygen concentration at which the first exact maximum occurred. An AUC would be undefined if only one ploidy class were present at an oxygen concentration, although both classes were present at every concentration in the reported grid. Because the parameter values and ploidy classes came from the same post-fit ensemble, these AUCs are descriptive single-parameter separation measures rather than hypothesis tests, causal effects, independently evaluated prediction accuracy, or multivariable parameter-importance estimates.

Separate *in vivo* parameter-landscape embedding and exploratory clustering. For Figure 4B, the differential-evolution initial population was deterministically reconstructed for each of the 500 separate *in vivo* fits from its optimizer seed, ordered 20-variable parameter table, transformed bounds, and effective initialization rule. Although the saved configuration recorded `de_init_mode=hybrid`, the single-stage backend explicitly used uniform initialization; all 500 fit logs recorded `init_mode=uniform`, zero local candidates, and 255 uniform candidates after the parameter-table initializer. The full 20-variable population was reconstructed before feature selection to preserve the optimizer random-number sequence. Candidate 1, which reproduced the parameter-table initializer, was then removed from every 256-member population. The observation/calibration variables ρ_{2N} and σ_{burden} were excluded, leaving 18 mechanistic parameters and 127,500 retained initial-population rows. These were combined with the 500 best endpoints, giving 128,000 *in vivo*-only parameter vectors. Each parameter was transformed on its configured natural or $\log_{10}$ optimizer scale and z-scored across the combined data. The vectors were embedded in two dimensions by t-SNE using random seed 123, perplexity 30, Barnes–Hut approximation parameter $\theta = 0.5$, and 1,000 iterations.

The 500 best endpoints were retained and k-means clustering was applied to their two-dimensional t-SNE coordinates for $k = 2, \dots, 8$. The value maximizing the average silhouette coefficient was selected; this procedure selected $k = 3$ with average silhouette 0.655 and cluster sizes of 43, 412, and 45 solutions. For the cluster-wise parameter summaries aligned with the AUC heatmap, each natural-scale value was transformed on its configured optimizer scale and mapped to the unit interval defined by its configured transformed lower and upper bounds. The plotted prior-referenced position was $z_{\text{prior}} = (u - 0.5)/\sqrt{1/12}$, where u is the bound-scaled value; zero therefore denotes the midpoint of the configured interval and one unit is the standard deviation of a uniform distribution over that interval. Cluster summaries show the median, interquartile range, and 5th–95th percentiles within each cluster. To rank the degree of descriptive cluster separation, η_{cluster}^2 was calculated for each parameter as the between-cluster sum of squares divided by the total sum of squares on the prior-referenced scale; the maximum absolute pairwise difference between cluster means was also recorded. All 18 displayed parameters were inputs to this embedding and cluster construction. Because the clusters and displayed parameters were derived from the same fitted ensemble, cluster-associated parameter differences are descriptive fitted-landscape structure rather than independent validation, biological subtype evidence, or causal parameter effects.

Pooled parameter-landscape embedding and joint-fit warm-start selection. A separate pooled landscape was constructed from optimizer population samples labeled as initial points and the final fitted solutions from the separate *in vivo* and *in vitro* calibrations. This landscape was used to design the joint-fitting warm starts shown in Figure 5B; it is a joint-fitting initialization result rather than an embedding of optimized joint-fit endpoints, and it is distinct from the 18-parameter *in vivo*-only landscape above. Fourteen parameters shared by the two contexts were used: λ_{max} , $p_{\text{mis,base}}$, p_{misseg} , $k_{o,\text{mis}}$, s_{max} , β_{buf} , n_{exp} , p_{wgd} , α_{O_2} , γ_{growth} , μ_{hp} , γ_{μ} , $\text{O}_{2,c}$, and n_{O} . Parameters were transformed on their configured natural or $\log_{10}$ scales and standardized jointly across the pooled data. By configuration, the first stored population row for each seed was removed in both contexts. In the *in vivo* export that row represents an optimizer initializer, whereas the *in vitro* export does not distinguish it from an ordinary sampled population row; this asymmetric provenance is a limitation of the warm-start landscape, not of the component likelihoods. The resulting 228,000 parameter vectors—127,500 *in vivo* population points, 99,500 *in vitro* population points, and 500 final solutions from each context—were embedded in two dimensions by t-SNE using random seed 123, perplexity 30, Barnes–Hut approximation parameter $\theta = 0.5$, and 1,000 iterations. K-means clustering of the 500 final *in vivo* positions over $k = 2, \dots, 8$ selected $k = 3$ with average silhouette 0.816 and sizes of 99, 385, and 16; these are the primary regions used for the six warm-start representatives described above.

Fixed-oxygen response-curve classification and reliability. For each final *in vivo* parameter set, dominant mean ploidy and spectral gap were evaluated from 0% to 5% O_2 in increments of 0.025 percentage points, giving 201 oxygen values per solution and 100,500 seed–oxygen evaluations. Each response curve was smoothed with a degree-2 Gaussian-family LOESS fit with span 0.20 under the versioned rule `loess_persistent.v1`. A smoothing spline with `spar=0.65` was used if LOESS failed or produced values outside the implemented plausibility envelope; if smoothing still failed, the raw finite values were retained. Finite differences of the smoothed curve were classified as positive, negative, or effectively zero using a tolerance equal to the larger of 10^{-6} and 10^{-4} times

the fitted ploidy range. Persistent segments were required to contain at least three grid intervals, span at least 2% of the oxygen range, and have amplitude at least the larger of 0.01 ploidy units and 3% of the fitted range. A terminal plateau required at least 10% of the oxygen range after the last persistent segment and a trailing amplitude no greater than the segment-amplitude threshold. Curves with a fitted range no greater than 0.05 were classified as approximately flat. Remaining curves were classified from the ordered persistent slope signs as monotone increasing, monotone decreasing, a single increase or decrease followed by a plateau, U-shaped, inverted U-shaped, or complex nonmonotone. For a curve with multiple persistent signs to be reduced to a monotone class, the total reverse-direction amplitude could not exceed 5% of the larger of the forward amplitude and fitted range.

For seed-level reliability summaries, a response curve was flagged as unreliable when at least 25% of its oxygen-grid values had a spectral gap below 0.005. Curves were flagged for caution when any grid value had a gap below 0.005 or when at least 10% of grid values had a gap below 0.01; all remaining curves were labeled reliable. These diagnostics indicate weak separation between the first two eigendirections and are not additional fitted-data hypothesis tests.

For the pair-specific response surfaces in Figure 6B, the lowest retained joint MAP objective within each of the six warm-start pairs supplied one *in vivo* parameter vector. Each vector was evaluated on the same 201-point oxygen grid and on 60 log-spaced effective per-chromosome mis-segregation probabilities from 0.005 through 0.5, giving 12,060 operator evaluations per pair and 72,360 in total. To isolate this probability as a standardized input axis, $p_{\text{mis},\text{base}}$ was set to the grid value and the stress-linked increment p_{misseg} was set to zero; all other parameters retained the selected winner’s fitted values. This construction makes the per-chromosome missegregation probability constant across chromosome states at each grid point. The realized population-average probability was recalculated from the dominant eigenvector as $\sum_N v_N p_{\text{mis}}(N, O_2)$ and was required to agree with the requested input to within 10^{-8} . Dominant mean ploidy colored the response surface, and the white dotted contour marked locations with spectral gap below 0.005. Separately, the black trajectory for each panel was obtained from the unmodified fitted parameter vector; its vertical coordinate was the model-implied population-average probability $\sum_N v_N p_{\text{mis}}(N, O_2)$. Consequently, the colored grid is a post-fit standardized response surface, whereas the black curve identifies the portion of that plane traversed by the selected fitted function.

Embedding of oxygen-response classes and fit objectives. For the response-class diagnostic, the 500 final *in vivo* coordinates were taken from the pooled 14-parameter joint-warm-start t-SNE embedding described immediately above; no second t-SNE embedding was estimated. Figure 6C therefore retains that correct pooled coordinate system and does not use the *in vivo*-only 18-parameter Figure 4B coordinates. These endpoints were joined by optimizer seed to their smoothed fixed- O_2 response classes and complete fitted MAP objective values. Solutions at or below the empirical 10th percentile of the 500 objectives were highlighted, giving 50 highlighted solutions. The class and objective overlays were used descriptively to assess landscape overlap, not as independent tests of class separation.

Visualization and reporting. For visualization and reporting in cell-number units, observed tumor volume (Eq. (24)) is converted to an approximate cell-count range using the specified diploid

packing-density bounds:

$$N_{ij}^{\text{obs,low}} = V_{ij}^{\text{obs}} \rho_{2N,\text{min}}, \quad N_{ij}^{\text{obs,mid}} = V_{ij}^{\text{obs}} \sqrt{\rho_{2N,\text{min}} \rho_{2N,\text{max}}}, \quad N_{ij}^{\text{obs,high}} = V_{ij}^{\text{obs}} \rho_{2N,\text{max}}. \quad (57)$$

This conversion is used for plotting/interpretation only and is not part of the fitted likelihood, which is evaluated directly in the volume domain via Eq. (27).

1.4 Parameter Tables

The active fitted parameters and their calibration bounds are summarized in Table 5; fixed numerical quantities are summarized in Table 6. Values and bounds are reported on the natural scale. The bounds encode the configuration of the reported analyses and should not be interpreted as likelihood-based confidence intervals. Some ranges were informed by published biological scales, whereas parameters without reliable direct measurements were assigned engineering bounds for numerical calibration.

For the reported joint fits, the *in vivo*/*in vitro* bounds for each overlapping active biological parameter are merged into the joint union interval in Eq. (47), and all 14 such parameters are represented by the center-delta construction in Eq. (48). Parameters used only by one context retain their component-specific bounds.

Supplementary Table 5: Active fitted parameters and natural-scale bounds used by the separate and joint analyses reported here. A dash means that the parameter is not active in that component. For the 14 overlapping biological parameters, the joint admissible interval is the union of the separate *in vivo* and *in vitro* intervals. These bounds are calibration constraints, not confidence intervals.

Parameter	<i>In vivo</i> bounds	<i>In vitro</i> bounds	Joint bounds	Role in the reported model
λ_{max}	0.005–0.300	0.005–1.500	0.005–1.500	Maximum division rate (day^{-1}); soft-coupled.
$p_{\text{mis,base}}$	10^{-6} –0.100	10^{-6} –0.005	10^{-6} –0.100	Baseline per-chromosome missegregation probability; soft-coupled.
p_{misseg}	0.005–0.500	0.005–0.400	0.005–0.500	Maximal stress-induced missegregation increment; soft-coupled.
$k_{o,\text{mis}}$	0.001–0.020	10^{-5} –0.020	10^{-5} –0.020	Death-hazard half-saturation scale for stress-induced missegregation (day^{-1}); soft-coupled.
s_{max}	0–1	0–1	0–1	Maximum per-copy post-missegregation survival factor; soft-coupled.
β_{buf}	0.01–10	0.01–10	0.01–10	Ploidy-dependent post-missegregation viability-loss strength; soft-coupled.
n_{exp}	0.1–10	0.1–10	0.1–10	Exponent controlling ploidy dependence of post-missegregation survival; soft-coupled.
p_{wgd}	10^{-5} – 10^{-3}	10^{-5} – 10^{-3}	10^{-5} – 10^{-3}	Constant per-division WGD probability; soft-coupled.
α_{O_2}	0.5–4.0	0.5–4.0	0.5–4.0	Strength of resource-stress-dependent growth damping; soft-coupled.
γ_{growth}	1.5–4.5	1.5–4.5	1.5–4.5	Chromosome-number dependence of growth damping; soft-coupled.
μ_{hp}	0.005–0.300	0.005–0.300	0.005–0.300	Stress-associated death scale (day^{-1}); soft-coupled.
γ_{μ}	1.2–3.5	1.2–3.5	1.2–3.5	Chromosome-number penalty on stress-associated death; soft-coupled.
$O_{2,c}$	0.1–2.5	10^{-4} –2.5	10^{-4} –2.5	Critical level for the oxygen-linked resource-stress function (% O_2); soft-coupled.

Parameter	<i>In vivo</i> bounds	<i>In vitro</i> bounds	Joint bounds	Role in the reported model
n_O	0.5–5.0	10^{-4} –5.0	10^{-4} –5.0	Hill exponent for the oxygen-linked resource-stress function; soft-coupled.
S_0	2.0–5.0	–	<i>In vivo</i> only	Low-burden effective oxygen supply level (% O ₂).
κ_O	0.1–1.2	–	<i>In vivo</i> only	Oxygen-drop amplitude in the supply-demand target.
η	1.0–2.0	–	<i>In vivo</i> only	Chromosome-number-weighted oxygen-demand exponent.
ρ_{2N}	5.0×10^4 – 5.0×10^5	–	<i>In vivo</i> only	Diploid-cell density (cells mm ⁻³).
k_{clear}	0.005–0.300	–	<i>In vivo</i> only	Dead-biomass clearance rate (day ⁻¹).
σ_{burden}	0.100–0.150	–	<i>In vivo</i> only	Log-scale burden observation standard deviation.
σ_{growth}	–	0.001–1.000	<i>In vitro</i> only	Growth-rate observation standard deviation.
σ_{kary}	–	0.01–10.0	<i>In vitro</i> only	Chromosome-number observation smoothing/error scale.
$\mu_{2N,0}$	–	45.725–49.725	<i>In vitro</i> only	Initial near-diploid Gaussian chromosome-count mean.
$\sigma_{2N,0}$	–	0.5–3.0	<i>In vitro</i> only	Initial near-diploid Gaussian chromosome-count SD.
$\mu_{4N,0}$	–	83.667–91.667	<i>In vitro</i> only	Initial near-tetraploid Gaussian chromosome-count mean.
$\sigma_{4N,0}$	–	1.5–6.0	<i>In vitro</i> only	Initial near-tetraploid Gaussian chromosome-count SD.

Supplementary Table 6: Fixed numerical settings used by the reported resource-stress supply-demand model and calibration workflow. Values are reported on the natural scale; settings for inactive optional modules are identified explicitly.

Parameter	Value	Unit	Description	Reference
τ_{O_2}	0.1	day	Oxygen relaxation timescale.	Fixed numerical relaxation setting; empirical workflow choice rather than a literature-derived parameter.
$O_{2,\text{cap}}$	5.0	% O ₂	Maximum oxygen level used to cap the supply-demand oxygen trajectory, inherited from the S_0 upper bound.	Bertout et al. (2008); ²⁸
$O_{2,\text{min}}$	0	% O ₂	Lower oxygen floor used in the logarithmic oxygen target relation.	McKeown (2014). ²⁹
N_{ref}	10^6	cells	Reference viable burden scale used in the oxygen supply-demand relation.	Physical/numerical anoxia floor used to bound oxygen values; model constraint rather than literature-estimated threshold.
N_{unit}	22	autosomal chromosomes	Modeled autosomal chromosome count corresponding to ploidy 1.	Oxygen-feedback normalization constant; empirical model scaling value rather than a measured biological quantity.
N_{min}	22	autosomal chromosomes	Minimum autosomal chromosome-count state retained in the discrete state space.	Model convention defining one haploid autosome set as ploidy 1; definition rather than literature-estimated parameter.
N_{max}	154	autosomal chromosomes	Maximum autosomal chromosome-count state retained in the discrete state space.	Discrete state-space lower bound anchored to the model ploidy convention; grid-design choice.
Δt	0.05	day	Simulation time step.	Bielski et al. (2018); ¹ Zack et al. (2013). ³⁰
K	10^{12}	cells	Carrying-capacity setting for the optional logistic crowding term; crowding was disabled in the reported fits.	Simulation time step selected for numerical resolution and stability; computational setting rather than biological parameter.
$N(0)$	10^6	cells	Initial total tumor population size.	Waclaw et al. (2015). ³¹
σ_{ploidy}	0.12	chromosomes	Fixed endpoint autosomal chromosome-number observation-noise scale in the harvest likelihood.	Initial-condition scale for the simulation/calibration workflow; empirical setup choice.
λ_{nec}	1	unitless	Relative weight of the terminal necrosis loss in the <i>in vivo</i> objective (<code>lambda_necrosis</code> in the run configuration).	Fixed likelihood observation-noise term; empirical measurement-error setting in the calibration workflow.
σ_{nec}	0.75	logit-fraction units	Scale used to standardize terminal necrosis discrepancies (<code>sigma_necrosis_logit</code> in the run configuration).	Reported calibration setting.
ϵ_{nec}	10^{-4}	fraction	Clipping constant for observed and predicted necrosis fractions before the logit transform (<code>necrosis_fraction_eps</code> in the run configuration).	Reported calibration setting.
λ_{prior}	0.03	unitless	Relative weight of the soft-prior contribution to the <i>in vivo</i> objective.	Numerical safeguard in the reported calibration.

Parameter	Value	Unit	Description	Reference
$\varepsilon_{\log B}$	10^{-12}	mm^3	Volume-domain floor used inside $\log\{\max(V, \varepsilon_{\log B})\}$ for burden comparisons.	Numerical floor preventing $\log(0)$; computational safeguard.
$N_{\min, \text{pop}}$	10^{-12}	cells	Minimum population floor used for numerical stability.	Population floor used to avoid numerical underflow or zero states; computational safeguard.

Supplementary Table 7: Function-level contrasts reconstructed from all 3,000 joint-fit endpoints. Values are first summarized within each of the six warm-start pairs; ranges below span the six pair medians. Ratios are *in vivo*/*in vitro*. Numerical optimizer endpoints are not biological replicates or confidence-interval samples.

Derived quantity		Reference state	<i>In vivo</i> pair-median range	<i>In vitro</i> pair-median range	Direction across pairs
Per-copy survival	post-missegregation	$N = 44$	0.728–0.952	0.204	higher <i>in vivo</i> , 6/6
Per-copy survival	post-missegregation	$N = 88$	0.901–0.958	0.837	higher <i>in vivo</i> , 6/6
Survival gradient, $s_{88} - s_{44}$		$44 \rightarrow 88$	0.006–0.186	0.633	larger <i>in vitro</i> , 6/6
Complete missegregation-probability ratio	effective	0% O ₂ , $N = 44$		11.140–48.086	> 1 , 6/6
Complete missegregation-probability ratio	effective	0% O ₂ , $N = 88$		11.185–47.762	> 1 , 6/6
Complete missegregation-probability ratio	effective	1% O ₂ , $N = 44$		7.426–29.128	> 1 , 6/6
Complete missegregation-probability ratio	effective	1% O ₂ , $N = 88$		10.361–42.923	> 1 , 6/6
Complete missegregation-probability ratio	effective	5% O ₂ , $N = 44$		0.389–5.448	> 1 , 3/6
Complete missegregation-probability ratio	effective	5% O ₂ , $N = 88$		2.321–5.697	> 1 , 6/6

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